

VOJTA'S ABC CONJECTURE FOR ENTIRE CURVES IN TORIC VARIETIES HIGHLY RAMIFIED OVER THE BOUNDARY

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ABSTRACT. We prove Vojta's abc conjecture for projective space $\mathbb{P}^n(\mathbb{C})$, assuming that the entire curves in $\mathbb{P}^n(\mathbb{C})$ are highly ramified over the coordinate hyperplanes. This extends the results of Guo Ji and the second-named author for the case $n = 2$ (see [9]). We also explore the corresponding results for projective toric varieties. Consequently, we establish a version of Campana's orbifold conjecture for finite coverings of projective toric varieties.

1. INTRODUCTION

Let X be an algebraic variety and D be an effective divisor (possibly empty) on X . The set $X \setminus D$ is *Brody hyperbolic* if every entire curve $f : \mathbb{C} \rightarrow X \setminus D$ is constant. It is *Brody quasi-hyperbolic* if every entire curve $f : \mathbb{C} \rightarrow X \setminus D$ is degenerate, meaning its image is contained a proper Zariski-closed subset Z of X . If such a subset Z can be chosen independently of f , then $X \setminus D$ is said to be *Brody strong quasi-hyperbolic* and Z is called an *exceptional set*. This leads us to the Griffiths-Lang's conjecture.

Griffiths-Lang's conjecture. *Let X be a smooth complex projective variety, D a normal crossing divisor on X , and K_X a canonical divisor on X . If $K_X + D$ is big, then $X \setminus D$ is Brody quasi-hyperbolic.*

A more precise quantitative version, in terms of the Nevanlinna's functions, can be formulated as follows.

Griffiths' conjecture. *For a holomorphic map $f : \mathbb{C} \rightarrow X$ with Zariski dense image, the inequality*

$$T_{K_X+D,f}(r) \leq_{\text{exc}} N_f(D, r) + \epsilon T_{A,f}(r)$$

holds for any $\epsilon > 0$, where A is an ample divisor on X , and the notation $\leq_{\text{exc}}$ means the inequality holds for all $r \in (0, \infty)$ except for a set of finite measure.

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Here, for each positive integer n , $N_f^{(n)}(D, r)$ is the n -truncated counting function with respect to D given by

$$N_f^{(n)}(D, r) = \sum_{0 < |z| < r} \min\{\text{ord}_z f^* D, n\} \log \frac{r}{|z|} + \min\{\text{ord}_0 f^* D, n\} \log r,$$

$N_f(D, r)$ is the counting function with respect to D given by $N_f^{(\infty)}(D, r)$, and $T_{D,f}(r)$ is the (Nevanlinna) height function relative to the divisor D (referring to [27, Section 12]).

Paul Vojta formulated the following general abc conjecture (see [27, Conjecture 15.2] and [27, Conjecture 23.4]), which extends the above Griffiths' conjecture. It is also viewed as the quantitative version of Campana's orbifold conjecture which will be stated later in the introduction.

Vojta's abc conjecture. *Let X be a smooth complex projective variety, D a normal crossing divisor on X , and K_X a canonical divisor on X . For any $\epsilon > 0$, there exists a proper Zariski-closed subset Z of X such that for any entire curve $f : \mathbb{C} \rightarrow X$ whose image is not contained in Z , the inequality*

$$T_{K_X+D,f}(r) \leq_{\text{exc}} N_f^{(1)}(D, r) + \epsilon T_{A,f}(r)$$

holds for any $\epsilon > 0$, where A is an ample divisor on X .

Note that Vojta's abc conjecture strengthens Griffiths' conjecture in two key ways: (i) the exceptional set Z above is independent of the map f , and (ii) the counting function is truncated at level one. Proving these two features is quite challenging, even if the aforementioned version of Griffiths' conjecture could be established. In this paper, we first prove Vojta's abc conjecture for entire curves in $\mathbb{P}^n(\mathbb{C})$, assuming the map is highly ramified over the coordinate hyperplanes. The precise statement of our main result is stated below. Additionally, we highlight that the exceptional set obtained in the theorem can be determined explicitly-this is a challenging issue in the study of Nevanlinna theory and complex hyperbolicity.

Main Theorem. *Let G be a non-constant homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_n]$ with no monomial factors and no repeated factors. Assume that $[G = 0]$ is in general position with the coordinate hyperplanes $H_i = [x_i = 0]$ in $\mathbb{P}^n(\mathbb{C})$, $0 \leq i \leq n$. Then, for any $\epsilon > 0$, there exist an effectively computable positive integer ℓ and a proper Zariski closed subset Z of $\mathbb{P}^n(\mathbb{C})$ that can be explicitly determined, such that the following two inequalities hold:*

- (i) $N_{G(\mathbf{f})}(0, r) - N_{G(\mathbf{f})}^{(1)}(0, r) \leq_{\text{exc}} \epsilon T_f(r)$, and
- (ii) $(\deg G - \epsilon) T_f(r) \leq_{\text{exc}} N_{G(\mathbf{f})}^{(1)}(0, r)$,

for any non-constant holomorphic map $f : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ whose image is not contained in Z , and where $\text{mult}_{z_0}(f^ H_i) \geq \ell$ for all i with $0 \leq i \leq n$ and for every $z_0 \in \mathbb{C}$ with $f(z_0) \in H_i$. Here,*

the evaluation of G is taken at $\mathbf{f} = (f_0, \dots, f_n)$ in a reduced form of f , meaning that $f_0, \dots, f_n$ are entire functions without common zeros.

We note that this theorem corresponds to a case of Vojta's abc conjecture for entire curves with $D = H_0 + \dots + H_n + [G = 0]$, so that $K_{\mathbb{P}^n+D}$ is linearly equivalent to $[G = 0]$. The case $n = 2$ was previously established by Ji Guo and the second named author ([9, Theorem 1.4]). The new idea in this paper is motivated by the recent paper [2], where the authors considered the parabolic Riemann surface $\mathcal{Y}_f := \mathbb{C} \setminus f^{-1}(D)$, reducing the problem to the omitting case. To achieve this in our setting, we need to extend the results in [7] and [9] to the parabolic setting.

We state some consequences of our Main Theorem, starting with the introduction of Campana's orbifolds. Instead of considering the complement case in Brody's hyperbolicity, we can also examine an entire curve $f : \mathbb{C} \rightarrow X$ that is ramified over D . According to Campana [1], an *orbifold divisor* is defined as $\Delta := \sum_{Y \subset X} (1 - m_\Delta^{-1}(Y)) \cdot Y$, where Y ranges over all irreducible divisors of X , and $m_\Delta(Y) \in [1, \infty) \cap \mathbb{Q}$, with all but finitely many values equal to one. An *orbifold entire curve* $f : \mathbb{C} \rightarrow (X, \Delta)$, with $\Delta = \sum_i (1 - m_i^{-1})D_i$, is a homomorphic map such that $f(\mathbb{C}) \not\subset \text{Supp}(\Delta)$ and $\text{mult}_{z_0}(f^*D_i) \geq m_i$ for all i and for every $z_0 \in \mathbb{C}$ with $f(z_0) \in D_i$.

Campana's orbifold conjecture. *If (X, Δ) is an orbifold pair such that $K_X + \Delta$ is big, then there exists a proper closed subvariety $Z \subset X$ containing all the non-constant orbifold entire curves $f : \mathbb{C} \rightarrow (X, \Delta)$.*

Remarks.

- (i) The Campana's conjecture is a consequence of the above Vojta's abc conjecture.
- (ii) Given an orbifold divisor Δ on X , let $D := \text{Supp}(\Delta) = D_1 + \dots + D_q$. Recall that the pair (X, D) is said to be of *log-general type* if the divisor $K_X + D$ is big. It is clear that (X, D) is of log-general type if (X, Δ) is of general type, i.e. $K_X + \Delta$ is big. On the other hand, the condition that (X, D) is of log-general type implies that (X, Δ) is of general type if (X, Δ) has a sufficiently large multiplicity along each D_i , $1 \leq i \leq q$. (See [11, Corollary 2.2.24].)

The Main Theorem leads to the following two theorems concerning Campana's orbifold conjecture; however, we will not include the proofs, as they can be adapted from those in [9] for the case $n = 2$. We begin with the result concerning $n + 2$ components.

Theorem 1. *Let Δ_0 be an orbifold divisor on $\mathbb{P}^n(\mathbb{C})$ and let $H_0, H_1, \dots, H_n$ be the coordinate hyperplanes in $\mathbb{P}^n(\mathbb{C})$, such that Δ_0 and $H_0, H_1, \dots, H_n$ are in general position. Let $m_i \in (1, \infty) \cap \mathbb{Q}$, $0 \leq i \leq n$, and $\Delta = \Delta_0 + (1 - \frac{1}{m_0})H_0 + (1 - \frac{1}{m_1})H_1 + \dots + (1 - \frac{1}{m_n})H_n$. Assume that $\deg \Delta > n + 1$. Then there exist a proper Zariski closed subset W of $\mathbb{P}^n(\mathbb{C})$ and an effectively*

computable positive integer ℓ such that the image of any orbifold entire curve $f : \mathbb{C} \rightarrow (\mathbb{P}^n(\mathbb{C}), \Delta)$ with $\min\{m_0, m_1, \dots, m_n\} \geq \ell$ must be contained in W .

The next case involves $n + 1$ components, not all are hyperplanes, in $\mathbb{P}^n(\mathbb{C})$.

Theorem 2. *Let F_i , $1 \leq i \leq n + 1$, be homogeneous irreducible polynomials of positive degrees in $\mathbb{C}[x_0, \dots, x_n]$. Assume that $D_i := [F_i = 0] \subset \mathbb{P}^n(\mathbb{C})$, $1 \leq i \leq n + 1$, intersect transversally. Let $\Delta = (1 - \frac{1}{m_1})D_1 + \dots + (1 - \frac{1}{m_{n+1}})D_{n+1}$ be an orbifold divisor of $\mathbb{P}^n(\mathbb{C})$. Assume that $\deg \Delta > n + 1$. Then, there exist an effectively computable positive integer ℓ and a proper Zariski closed subset W of $\mathbb{P}^n(\mathbb{C})$ such that the image of any orbifold entire curve $f : \mathbb{C} \rightarrow (\mathbb{P}^n(\mathbb{C}), \Delta)$ with $\min\{m_1, \dots, m_{n+1}\} \geq \ell$ must be contained in W .*

Note that $\mathbb{P}^n(\mathbb{C})$ is a toric variety when we identify $\mathbb{G}_m^n$ with $\mathbb{P}^n(\mathbb{C}) \setminus \cup_{i=0}^n H_i$, where $H_0, \dots, H_n$ are the $n+1$ coordinate hyperplanes. A toric variety X is defined as a variety over $\mathbb{C}$ that contains $\mathbb{G}_m^n$ as a dense open subvariety, with the action of $\mathbb{G}_m^n$ on itself extends to an algebraic action of $\mathbb{G}_m^n$ on X . We refer to [5] for more details on toric varieties. Using the arguments in [12] and [6], we can extend our Main Theorem to projective toric varieties. Recall from [12] that an “admissible pair” is a couple (X, V) where V is a nonsingular variety embedded in a nonsingular projective variety X in such a way that $D_0 = X \setminus V$ is a normal crossings divisor. A Campana orbifold (X, Δ) is said to be *associated to the admissible pair* (X, V) if D_0 is the support of Δ .

Theorem 3. *Let X be a nonsingular projective toric variety and let D be an effective reduced divisor on the admissible pair $(X, \mathbb{G}_m^n)$, such that D is in general position with the boundary $D_0 = X \setminus \mathbb{G}_m^n$. Let A be a big divisor on X . Then, for every $\epsilon > 0$, there is a positive integer ℓ and a proper Zariski-closed subset Z of X such that, for any Campana orbifold (X, Δ) associated to the admissible pair $(X, \mathbb{G}_m^n)$ with multiplicities at least ℓ along the boundary D_0 , the following two inequalities hold:*

- (i) $N_f(D, r) - N_f^{(1)}(D, r) \leq_{\text{exc}} \epsilon T_{A,f}(r)$, and
- (ii) $T_{D,f}(r) \leq_{\text{exc}} N_f^{(1)}(D, r) + \epsilon T_{A,f}(r)$

for any non-constant entire orbifold curve f on (X, Δ) such that the image of f is not contained in Z .

Theorem 3 implies Campana’s conjecture for an orbifold pair that is a finite covering of a toric variety, as follows.

Theorem 4. *Let X be a nonsingular projective toric variety, and let $D_0 = X \setminus \mathbb{G}_m^n$. Let Y be a nonsingular complex projective variety with a finite morphism $\pi : Y \rightarrow X$. Let $H := \pi^*(D_0)$ and $R \subset Y$ the ramification divisor of π omitting components from the support of H . Assume*

that $\pi(R)$ and D_0 are in general position on X . Then there exists a positive integer ℓ such that, if the orbifold pair (Y, Δ) is of general type with $\text{Supp}(\Delta) = \text{Supp}(H)$ and multiplicity at least ℓ for each component, then there exists a proper Zariski closed subset W of Y such that the image of any non-constant orbifold entire curve $f : \mathbb{C} \rightarrow (Y, \Delta)$ must be contained in W .

The paper is organized as follows. In Section 2, we prove a version of the Main Theorem for rational maps and explain the motivation for using parabolic Riemann surfaces. Section 3 reviews basic results from Nevanlinna theory relevant to parabolic settings, as well as the key theorems from [2]. Furthermore, we extend the GCD theorem developed in [13] and in [9] to parabolic setting. In Section 4, we establish the abc theorem for parabolic Riemann surfaces, building on the methods developed in [7]. Section 5 presents the proof of the Main Theorem, while Section 6 extends results to toric varieties.

2. MOTIVATION THROUGH THE RATIONAL CASE

In [7], a version of Vojta's abc conjecture for algebraic tori over algebraic function fields is established. As we will see, the algebraic reduction method presented in [7] is a key technical component in proving our Main Theorem. The core idea behind the proof is also inspired by the function field approach. In this section, we will prove a version of the Main Theorem for rational functions and explain the role of parabolic Riemann surfaces in this context.

2.1. The Rational Case. The statements and arguments apply verbatim to algebraic function fields (see also [6, Theorem 1.13]). Here, let $K = \mathbb{C}(t)$ be the function field of $\mathbb{P}^1(\mathbb{C})$. Let S be a finite subset of $\mathbb{P}^1(\mathbb{C})$. We denote by $\mathcal{O}_S^*$ the set of S -units, which consists of rational functions with no zeros and poles outside of S . For a rational function $f = \frac{P}{Q}$, where P and Q are coprime polynomials, we define $\deg f := \max\{\deg P, \deg Q\}$ and denote by $\text{ord}_{\mathbf{p}}^+(f)$ the order of the zero of f at $\mathbf{p} \in \mathbb{P}^1(\mathbb{C})$. We use the notation $N_S(f)$ (respectively $N_S^{(1)}(f)$) to count the number of zeros of f with multiplicity (respectively, without multiplicity) outside of S . When S is empty, we simply write $N(f)$, which equals $\deg f$, to count the number of zeros with multiplicity on $\mathbb{P}^1(\mathbb{C})$; $N^{(1)}(f)$ denotes the same count without multiplicity.

Recall a simplified version of the main technical theorem in [7].

Theorem 5 ([7, Theorem 4]). *Let $G \in \mathbb{C}[x_1, \dots, x_n]$ be a non-constant homogeneous polynomial with neither monomial factors nor repeated factors. Let S be a finite subset of $\mathbb{P}^1(\mathbb{C})$. Then, for any $\epsilon > 0$, there exist a positive real number c_0 and a proper Zariski closed subset Z of $\mathbb{A}^n$ such that, for all $\mathbf{u} := (u_1, \dots, u_n) \in (\mathcal{O}_S^*)^n \setminus Z$, we have either*

(i) *the inequality*

$$\max_{1 \leq i \leq n} \{\deg(u_i)\} \leq c_0 \max\{1, |S| - 2\}$$

holds,

(ii) *or the following two statements hold*

(a) $N_S(G(\mathbf{u})) - N_S^{(1)}(G(\mathbf{u})) \leq \epsilon \max_{1 \leq i \leq n} \{\deg u_i\}$ if $G(u_1, \dots, u_n) \neq 0$.

(b) If $G(0, \dots, 0) \neq 0$ and $\deg_{X_i} G = \deg G = d$ for $1 \leq i \leq n$, then

$$N_S^{(1)}(G(\mathbf{u})) \geq (\deg G - \epsilon) \max_{1 \leq i \leq n} \{\deg u_i\}.$$

Here c_0 can be effectively bounded from above in terms of ϵ , n , and the degree of G . Moreover, the exceptional set Z can be expressed as the zero locus of a finite set $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$ with the following properties: (Z1) Σ depends on ϵ and G but independent of S and can be determined explicitly and (Z2) $|\Sigma|$ and the degree of each polynomial in Σ can be effectively bounded from above in terms of ϵ , n , and the degree of G .

The construction of the set of polynomials Σ follows from the specialization lemmas [7, Lemma 15 and 16]. In particular, it is evident that Σ is independent of the choice of S . Note that $\mathbf{u} := (u_1, \dots, u_n) \in (\mathcal{O}_S^*)^n \setminus Z$ means that $F(\mathbf{u}) \neq 0$ (a nontrivial function and hence not identically zero) for every $F \in \Sigma$. This formulation is closely analogous to the number field case. Alternatively, one may view $\mathbf{u}$ as a morphism: $\mathbb{P}^1(\mathbb{C}) \setminus S \rightarrow \mathbb{C}^n$; then the condition $F(\mathbf{u}) \neq 0$ for every $F \in \Sigma$ is equivalent to saying that the image $\mathbf{u}(\mathbb{P}^1(\mathbb{C}) \setminus S)$ is not contained in Z .

We will demonstrate how to apply the above theorem to show the following version of Main Theorem for rational case.

Theorem 6. *Let G be a non-constant homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_n]$ with no monomial factors and no repeated factors. Assume $[G = 0]$ is in general position with the coordinate hyperplanes $H_i = [x_i = 0]$, $0 \leq i \leq n$. Then, for any $\epsilon > 0$, there exist effectively computable positive integers ℓ and a proper Zariski closed subset Z of $\mathbb{P}^n(\mathbb{C})$ such that the following two inequalities hold:*

(i) $N(G(\mathbf{f})) - N^{(1)}(G(\mathbf{f})) \leq \epsilon \deg \mathbf{f}$, and

(ii) $N^{(1)}(G(\mathbf{f})) \geq (\deg G - \epsilon) \deg \mathbf{f}$,

for any non-constant entire rational curve $\mathbf{f} = (f_0, \dots, f_n)$, where $f_0, \dots, f_n \in \mathbb{C}[z]$ are polynomials without common zeros, provided that $\mathbf{f}(\mathbb{C}) \not\subset Z$ and that the following condition is satisfied: for each $0 \leq i \leq n$ and $\mathbf{p} \in \mathbb{P}^1(\mathbb{C})$, if $\text{ord}_{\mathbf{p}}(f_i) > 0$, then $\text{ord}_{\mathbf{p}}(f_i) \geq \ell$. Here, $\deg \mathbf{f} = \max\{\deg f_0, \dots, \deg f_n\}$.

In addition to Theorem 5, we also require the following result to prove Theorem 6.

Proposition 7 ([7, Corollary 6]). *Let $G = \sum_{\mathbf{i} \in I_G} a_{\mathbf{i}} \mathbf{x}^{\mathbf{i}} \in \mathbb{C}[x_1, \dots, x_n]$ be a non-constant polynomial, where $a_{\mathbf{i}} \neq 0$ if $\mathbf{i} \in I_G$. Assume that $G(0, \dots, 0) \neq 0$. Let Z be the Zariski closed subset that is the union of hypersurfaces of $\mathbb{A}^n$ of the form $\sum_{\mathbf{i} \in I} a_{\mathbf{i}} \mathbf{x}^{\mathbf{i}} = 0$ where I is a non-empty subset of I_G . Let S be a finite subset of $\mathbb{P}^1(\mathbb{C})$. Then, for all $(u_1, \dots, u_n) \in (\mathcal{O}_S^*)^n \setminus Z$, we have*

$$\sum_{\mathbf{p} \in S} \text{ord}_{\mathbf{p}}^+(G(u_1, \dots, u_n)) \leq \tilde{c} \max\{1, |S| - 2\}$$

where $\tilde{c} = \frac{1}{2} \binom{n+\deg G}{n} (\binom{n+\deg G}{n} + 1)$.

The proof of Theorem 6 (i) can be outline as the follows. Given a rational map $\mathbf{f} = (f_0, \dots, f_n)$, where $f_0, \dots, f_n \in \mathbb{C}[z]$ are polynomials without common zeros, we define a set $S_{\mathbf{f}}$ consisting of the zeros of all f_i . With this choice, each coordinate function f_i becomes an $S_{\mathbf{f}}$ -unit, so that Theorem 5 (ii) can be applied to estimate the multiplicities of the zeros of $G(\mathbf{f})$ outside $S_{\mathbf{f}}$. We then use Proposition 7 to estimate the zeros of $G(\mathbf{f})$ inside $S_{\mathbf{f}}$. Crucially, the constants and exceptional sets appearing in Theorem 5 and Proposition 7 do not depend on S , allowing this method to be used for all rational maps.

Proof of Theorem 6. Denote by ℓ a sufficiently large integer to be determined later. Let $\mathbf{f} = (f_0, \dots, f_n)$, where $f_0, \dots, f_n \in \mathbb{C}[z]$ are polynomials without common zeros. By the assumption, for each $0 \leq i \leq n$ and $\mathbf{p} \in \mathbb{P}^1(\mathbb{C})$, if $\text{ord}_{\mathbf{p}}(f_i) > 0$, then $\text{ord}_{\mathbf{p}}(f_i) \geq \ell$. Let

$$(2.1) \quad \tilde{S}_{\mathbf{f}} := \{\mathbf{p} \in \mathbb{P}^1(\mathbb{C}) \mid \text{ord}_{\mathbf{p}}(f_i) > 0 \text{ for some } 0 \leq i \leq n\} \cup \{\infty\}.$$

Then

$$(2.2) \quad |\tilde{S}_{\mathbf{f}}| \leq \sum_{i=0}^n N_S^{(1)}(f_i) + 1 \leq \frac{n+1}{\ell} \max_{0 \leq i \leq n} \{\deg f_i\} + 1,$$

and $f_i \in \mathcal{O}_{\tilde{S}_{\mathbf{f}}}^*$ for $0 \leq i \leq n$. Let $\tilde{G}(x_1, \dots, x_n) := G(1, x_1, \dots, x_n)$ and $u_i = \frac{f_i}{f_0} \in \mathcal{O}_{\tilde{S}_{\mathbf{f}}}^*$, $1 \leq i \leq n$. Then $\tilde{G}(u_1, \dots, u_n) := f_0^{-\deg G} G(\mathbf{f})$. An elementary argument shows that

$$(2.3) \quad \max_{1 \leq i \leq n} \{\deg u_i\} \leq \max_{0 \leq i \leq n} \{\deg f_i\} \leq \sum_{j=1}^n \deg u_j.$$

Let $\epsilon > 0$ be given. We now apply Theorem 5 for $\tilde{G} \in \mathbb{C}[x_1, \dots, x_n]$ and $\tilde{S}_{\mathbf{f}}$ to find a positive integer c_0 and a proper Zariski closed subset Z_0 of $\mathbb{A}^n$ independent of $\tilde{S}_{\mathbf{f}}$ such that we have either the inequality

$$(2.4) \quad \max_{0 \leq i \leq n} \{\deg f_i\} (= \deg \mathbf{f}) \leq c_0 \max\{1, |\tilde{S}_{\mathbf{f}}| - 2\}$$

holds or the following two inequalities hold:

$$(2.5) \quad N_{\tilde{S}_{\mathbf{f}}}(G(\mathbf{f})) - N_{\tilde{S}_{\mathbf{f}}}^{(1)}(G(\mathbf{f})) = N_{\tilde{S}_{\mathbf{f}}}(\tilde{G}(u_1, \dots, u_n)) - N_{\tilde{S}_{\mathbf{f}}}^{(1)}(\tilde{G}(u_1, \dots, u_n)) \leq \epsilon \deg \mathbf{f},$$

$$(2.6) \quad N_{\tilde{S}_f}^{(1)}(G(\mathbf{f})) = N_{\tilde{S}_f}^{(1)}(\tilde{G}(u_1, \dots, u_n)) \geq (\deg G - \epsilon) \cdot \deg \mathbf{f},$$

if $\mathbf{f}(\mathbb{C}) \not\subset \tilde{Z}_0$, where $\tilde{Z}_0$ is the projective closure of Z_0 in $\mathbb{P}^n(\mathbb{C})$. Since the constant c_0 and the exceptional set Z in Theorem 5 do not depend on S , it is important to note that c_0 and $\tilde{Z}_0$ do not depend on $\mathbf{f}$ as well. We first show that (2.4) does not hold if $\ell > 2c_0n$. Indeed, if (2.4) holds with $\ell > 2c_0n$, then it follows from (2.2) and the choice of ℓ that (2.4) yields

$$\max_{0 \leq i \leq n} \{\deg f_i\} \leq \frac{(n+1)c_0}{\ell} \max_{0 \leq i \leq n} \{\deg f_i\} + c_0 < \frac{1}{2} \max_{0 \leq i \leq n} \{\deg f_i\} + c_0.$$

Therefore, $\max_{0 \leq i \leq n} \{\deg f_i\} < 2c_0$. This is not possible since the f_i are not all constant, and by our assumption that every zero has multiplicity at least ℓ , we have $2nc_0 < \ell < \max_{0 \leq i \leq n} \{\deg f_i\}$.

We continue to let $\ell > 2c_0n$. As we have shown that (2.4) does not hold, it follows that (2.5) and (2.6) both hold if $\mathbf{f}(\mathbb{C}) \not\subset \tilde{Z}_0$. Then it follows trivially from (2.6) that

$$(2.7) \quad N^{(1)}(G(\mathbf{f})) \geq N_{\tilde{S}_f}^{(1)}(G(\mathbf{f})) \geq (\deg G - \epsilon) \cdot \deg \mathbf{f}.$$

This shows (ii) holds.

It remains to show (i). We now compute zeros for points in $\tilde{S}_f$. We will use Proposition 7. Let W be the Zariski closed subset of $\mathbb{P}^n(\mathbb{C})$ containing all hypersurfaces defined by all possible subsums (including G) in the expansion of G . Assume that $\mathbf{f}(\mathbb{C}) \not\subset W$. Since $\tilde{G}(0, \dots, 0) = G(1, 0, \dots, 0) \neq 0$, we now apply Proposition 7 to get

$$\sum_{\mathbf{p} \in \tilde{S}_f} \text{ord}_{\mathbf{p}}^+(\tilde{G}(u_1, \dots, u_n)) \leq \tilde{c} \max\{1, |\tilde{S}_f| - 2\},$$

where $\tilde{c} = \frac{1}{2} \binom{n+\deg G}{n} (\binom{n+\deg G}{n} + 1)$. Together with (2.2) and that $\ell \leq \max_{0 \leq i \leq n} \{\deg f_i\}$, it yields

$$m_{\tilde{S}_f}(\tilde{G}(u_1, \dots, u_n)) := \sum_{\mathbf{p} \in \tilde{S}_f} \text{ord}_{\mathbf{p}}^+(\tilde{G}(u_1, \dots, u_n)) \leq \frac{(n+1)\tilde{c}}{\ell} \max_{0 \leq i \leq n} \{\deg f_i\} \leq \epsilon \deg \mathbf{f}$$

provided $\ell > (n+1)\tilde{c}\epsilon^{-1}$. Combining with (2.5), we have

$$(2.8) \quad N(\tilde{G}(u_1, \dots, u_n)) - N^{(1)}(\tilde{G}(u_1, \dots, u_n)) \leq 2\epsilon \deg \mathbf{f}.$$

We now repeat the above arguments for $(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i})$, $0 \leq i \leq n$. Then we find a Zariski closed subset Z and a positive constant c , independent of $\mathbf{f}$, such that for each $0 \leq i \leq n$ we have

$$(2.9) \quad N\left(G\left(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i}\right)\right) - N^{(1)}\left(G\left(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i}\right)\right) \leq 2\epsilon \deg \mathbf{f}.$$

Since $f_0, \dots, f_n$ are polynomials without common zeros, for each $z \in \mathbb{C}$ there exists at least one f_i such that $f_i(z) \neq 0$. Then

$$\text{ord}_z^+(G(f_0, \dots, f_n)) = \text{ord}_z\left(G\left(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i}\right)\right).$$

This shows that

$$\begin{aligned} N(G(\mathbf{f})) - N^{(1)}(G(\mathbf{f})) &\leq \sum_{i=0}^n \left(N \left(G \left(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i} \right) \right) - N^{(1)} \left(G \left(\frac{f_0}{f_i}, \dots, \frac{f_n}{f_i} \right) \right) \right) \\ &\leq 2(n+1)\epsilon \deg \mathbf{f}. \quad (\text{by (2.9)}) \end{aligned}$$

The assertion (i) can be derived by adjusting ϵ . □

2.2. Motivation for the proof of the Main Theorem. In Section 2.1, the key to proving Theorem 6 is the use of the set S (depending on the map) that contains all zeros and poles of each coordinate function of the map. The crucial observation is that all coordinate functions then become S -units, allowing us to apply Theorem 5. Importantly, the constants and the exceptional set obtained in Theorem 5 are independent of S , which makes the argument work uniformly for all rational maps.

To extend this viewpoint to the analytic setting, we let $\mathcal{E}_f$ denote the set of all zeros of f_i , $0 \leq i \leq n$, and define

$$\mathcal{Y}_f := \mathbb{C} \setminus \mathcal{E}_f,$$

where $f = (f_0, \dots, f_n)$ is a holomorphic map such that the entire functions f_i , $0 \leq i \leq n$, have no common zeros. Then $\mathcal{Y}_f$ is a parabolic open Riemann surface equipped with a smooth exhaustion function σ for which the smooth $(1,1)$ -form $dd^c \log \sigma$ has finite total mass on $\mathcal{Y}_f$. Although σ may not be a parabolic exhaustion for $\mathcal{Y}_f$, Chen et al. (see [2]) observed that, after appropriate modifications, the key ingredients of Nevanlinna theory still admit analogues in this setting.

3. NEVANLINNA THEORY AND THE GCD THEOREM ON PARABOLIC RIEMANN SURFACES

3.1. Parabolic Riemann surfaces and Jensen's formula. A non-compact Riemann surface $\mathcal{Y}$ is *parabolic* if it admits a smooth exhaustion function $\sigma : \mathcal{Y} \rightarrow [0, \infty)$ such that $\log \sigma$ is harmonic outside a compact subset of $\mathcal{Y}$. Such σ is called a *parabolic exhaustion*. Nevanlinna theory on parabolic open Riemann surfaces was recently developed by Păun and Sibony [16]. Chen et al. [2] considered the case when $\mathcal{Y} := \mathbb{C} \setminus \mathcal{E}$, where $\mathcal{E} := \{a_j\}_{j=1}^\infty$ is a discrete countable set of points in $\mathbb{C}$. The surface $\mathcal{Y}$ is parabolic due to [23, Lemma 10.1(g), p. 75] by taking $\phi : \mathcal{Y} \rightarrow \mathbb{C}$ as the restriction of a transcendental function sending $\mathcal{E}$ to 0. However the smooth exhaustion function $\sigma : \mathcal{Y} \rightarrow [0, \infty)$ constructed in [2], which serves as our purpose, is not necessarily compactly supported, although the smooth $(1,1)$ -form $dd^c \log \sigma$ is of finite total mass on $\mathcal{Y}$. Therefore the parabolic Nevanlinna theory developed by Păun and Sibony is no longer directly applicable to $\mathcal{Y}$. This difficulty arises because the parabolic Jensen formula (see (3.8)) acquires

an additional term $\int_{B_r^\sigma} v dd^c \tau$. A key observation is that, by shrinking the radii r_j sufficiently small, we can make (see Lemma 10) $c_{g,\sigma}^+(r) := \int_{B_r^\sigma} \log^+ |g| dd^c \tau$ to be bounded (independent of r), for any meromorphic function g on $\mathbb{C}$. The subtle point, however, is that the required shrinking depends on the particular function g , so we must ensure in our arguments that such shrinking occurs only finitely many times. This “shrinking trick” was already observed by Chen et al. in [2] and indeed plays a central role in their work. Unfortunately many of the necessary details were not made explicit there. We therefore supply the details in this and the subsequent sections.

Let us first briefly review the construction of σ given in Lemma 3.1 and 3.2 of [2]. Let $\mathcal{Y} := \mathbb{C} \setminus \mathcal{E}$, where $\mathcal{E} := \{a_j\}_{j=1}^\infty$ is a discrete countable set of points in $\mathbb{C}$. Up to translation and dilation on $\mathbb{C}$, we may assume that $|a_j| > 2$ for all $j \geq 1$. We also assume that the radii satisfy the following conditions: (i) $\mathbb{D}(a_j, 2r_j)$ for all j and the disc $\mathbb{D}_2$ are disjoint, and (ii) the sum $\sum_{j \geq 1} r_j < +\infty$.

The exhaustion function σ on $\mathcal{Y}$ is obtained by smoothing the piecewise smooth exhaustion function $\hat{\sigma}$ given by

$$(3.1) \quad \log \hat{\sigma} := \log^+ |z| + \sum_{j \geq 1} r_j \log^+ \frac{r_j}{|z - a_j|}.$$

Note that $\hat{\tau} := \log \hat{\sigma}$ takes values in $[0, +\infty)$, is continuous on $\mathcal{Y}$ and is smooth (indeed harmonic) outside the circle $S(0, 1) := \{|z| = 1\}$ and the disjoint circles $S(a_j, r_j) := \{|z - a_j| = r_j\}$. By the Poincaré-Lelong formula (see, e.g., [17]), it is clear that

$$dd^c \hat{\tau} = \frac{1}{2} \nu(0, 1) + \frac{1}{2} \sum_{j=1}^\infty r_j (\nu(a_j, r_j) - \delta_{a_j})$$

is a distribution of order 0 and locally of finite mass, where $\nu(a_j, r_j)$ is the Haar measure on the circle $S(a_j, r_j)$.

The smoothing process goes as follows (see [2, Appendix]): Let

$$h(r) = \begin{cases} 0 & \text{for } r \leq \frac{3}{4} \\ \frac{1}{4\pi} \frac{1}{1 + e^{\frac{r-1}{(r-1)^2 - 1/16}}} & \text{for } \frac{3}{4} < r < \frac{5}{4} \\ \frac{1}{4\pi} & \text{for } r \geq \frac{5}{4} \end{cases}$$

and

$$(3.2) \quad H(r) := 4\pi \int_0^r h(s - c) \frac{ds}{s},$$

where the constant $c \in (0, 1/2)$ is chosen such that H agrees with $\log^+ r$ outside $[\frac{1}{2}, \frac{3}{2}]$. Note that H is an absolute function independent of a_j . This function $H(r)$ is smooth of $\log^+ r$ with

$0 \leq H(r) - \log^+ r \leq \log^+ \frac{3}{2} < \frac{1}{2}$ and $H(r) = \log^+ r$ when $H(r) \geq \log \frac{3}{2}$. Define

$$(3.3) \quad \tau := H(|z|) + \sum_{j=1}^{\infty} r_j H\left(\frac{r_j}{|z - a_j|}\right), \quad \sigma = \exp(\tau).$$

Then, as stated in [2, Lemma 3.1], $\sigma \geq \hat{\sigma}$ and the difference $\sigma - \hat{\sigma}$ is supported on

$$(3.4) \quad \text{Supp}(\sigma - \hat{\sigma}) \subset U := \left(A\left(0, \frac{1}{2}, \frac{3}{2}\right) \setminus \mathcal{E} \right) \cup \left(\bigcup_{j=1}^{\infty} A\left(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j\right) \right),$$

where $A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j) := \{z \in \mathcal{Y} : \frac{1}{2}r_j \leq |z - a_j| \leq \frac{3}{2}r_j\}$ are pairwise disjoint annuli. For $z \notin \bigcup_{j=1}^{\infty} \overline{\mathbb{D}(a_j, \frac{3}{2}r_j)}$, $\log^+ \left| \frac{r_j}{z - a_j} \right| = H\left(\left| \frac{r_j}{z - a_j} \right|\right) = 0$ for each j . Thus $\tau(z) = H(|z|)$ and $\hat{\tau}(z) = \log^+ |z|$, which are equal when $\tau(z) \geq \log \frac{3}{2}$. Hence $\sigma = \hat{\sigma}$ when $\sigma \geq \frac{3}{2}$. We say that *such σ is associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$* by assuming that $\{r_j\}$ satisfy (i) and (ii) above. Note that σ is not necessarily a parabolic exhaustion function because $dd^c \log \sigma$ may not be compactly supported.

Denote by $B_r^\sigma := \{y \in \mathcal{Y} : \sigma(y) < r\}$ and by $S_r^\sigma := \{y \in \mathcal{Y} : \sigma(y) = r\}$. Denote by μ_r the measure induced by the differential $d^c \log \sigma|_{S_r^\sigma}$ and write $d\mu_r = d^c \log \sigma|_{S_r^\sigma}$. Note that, from the above discussion (see also [2]),

$$(3.5) \quad B_t^\sigma \subset \mathbb{D}_t \text{ for } t \geq 1, \text{ and } \mathbb{D}_t \setminus \left(\bigcup_{j=1}^{\infty} \mathbb{D}(a_j, \frac{3}{2}r_j) \right) \subset B_t^\sigma \text{ for } t \geq \frac{3}{2}.$$

Lemma 8 (Lemma 3.2 in [2]). *The smooth 2-form $dd^c \log \sigma$ defines an order 0 distribution of finite mass on $\mathcal{Y}$.*

Proof. We use the formula: in polar coordinates $z = a + re^{i\theta}$, for some $a \in \mathbb{C}$, and for a smooth function ϕ , one has

$$d^c \phi = \frac{1}{4\pi} \left(r \frac{\partial \phi}{\partial r} d\theta - \frac{1}{r} \frac{\partial \phi}{\partial \theta} dr \right).$$

The smooth 2-form $dd^c \log \sigma$ is supported on U , where U is given in (3.4). On the annulus $A(0, \frac{1}{2}, \frac{3}{2})$, in polar coordinate $z = re^{i\theta}$, by (3.2) one has (see [2, Appendix])

$$(3.6) \quad dd^c \tau = dd^c H(|z|) = d\left(\frac{r}{4\pi} \frac{\partial H(r)}{\partial r} d\theta\right) = d(h(r - c)d\theta) = h'(r - c) dr \wedge d\theta = O(1)dr \wedge d\theta.$$

Thus $dd^c \tau$ has finite mass on $A(0, \frac{1}{2}, \frac{3}{2})$. On the annulus $A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j)$, since $\mathbb{D}(a_j, 2r_j)$ for all j and the disc $\mathbb{D}_2$ are disjoint, we have similarly that $\tau = r_j H\left(\frac{r_j}{|z - a_j|}\right)$ on $A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j)$.

Therefore,

$$\begin{aligned}
 dd^c \tau &= dd^c \left(r_j H \left(\frac{r_j}{|z - a_j|} \right) \right) \\
 (3.7) \quad &= d \left(\frac{r}{4\pi} \frac{\partial r_j H \left(\frac{r_j}{r} \right)}{\partial r} d\theta \right) = \left(\frac{r_j}{r} \right)^2 h' \left(\frac{r_j}{r} - c \right) dr \wedge d\theta = O(1) dr \wedge d\theta.
 \end{aligned}$$

Since $\sum_{j=1}^{\infty} r_j < +\infty$, the support U is of finite Lebesgue measure, this proves the lemma. $\square$

We now derive the parabolic Jensen's formula in our setting. For any function $v : \mathcal{Y} \rightarrow (-\infty, +\infty)$ such that $dd^c v$ is of order 0, the parabolic Jensen's formula (see [16, Remark 3.2, p. 22]) is stated as

$$(3.8) \quad \int_1^r \frac{dt}{t} \int_{B_t^\sigma} dd^c v = \int_{S_r^\sigma} v d\mu_r - \int_{B_r^\sigma} v dd^c \tau \quad (r > 1).$$

Let g be meromorphic on $\mathbb{C}$. In (3.8), by letting $v := \log |g|$, and also using the Poincaré-Lelong formula, we get the following lemma.

Lemma 9 (Parabolic Jensen's formula). *Let g be a meromorphic function on $\mathbb{C}$. Then*

$$\int_{S_r^\sigma} \log |g| d\mu_r = N_{g,\sigma}(0, r) - N_{g,\sigma}(\infty, r) + c_{g,\sigma}(r),$$

where $N_{g,\sigma}(0, r) := \int_1^r \sum_{z \in B_t^\sigma} \max\{0, \text{ord}_z g\} \frac{dt}{t}$, $N_{g,\sigma}(\infty, r) := N_{\frac{1}{g}, \sigma}(0, r)$, and

$$(3.9) \quad c_{g,\sigma}(r) := \int_{B_r^\sigma} \log |g| dd^c \tau.$$

Denote by

$$(3.10) \quad c_{g,\sigma}^+(r) := \int_{B_r^\sigma} \log^+ |g| dd^c \tau.$$

Lemma 10. *Let g be a meromorphic function on $\mathbb{C}$. Let $\mathcal{Y} \subset \mathbb{C}$ be given with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Then we can choose r_j small enough such that there exists a positive constant C_g depending on g with $c_{g,\sigma}^+(r) \leq C_g$ for any $r > 0$.*

Proof. By using (3.5) for $t \geq 2$, and notice that $dd^c \tau$ is supported on U where U is given in (3.4), we have

$$\begin{aligned}
 c_{g,\sigma}^+(t) &= \int_{B_t^\sigma} \log^+ |g| dd^c \tau \leq \int_{\mathbb{D}_t \setminus \mathcal{E}} \log^+ |g| dd^c \tau \\
 (3.11) \quad &= \int_{A(0, \frac{1}{2}, \frac{3}{2})} \log^+ |g| dd^c \tau + \int_{\cup_{j \geq 1} A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j) \cap \mathbb{D}_t} \log^+ |g| dd^c \tau.
 \end{aligned}$$

Notice that $\log^+ |g|$ is locally integrable around the zeros or poles of g and by (3.6), we get

$$\int_{A(0, \frac{1}{2}, \frac{3}{2})} \log^+ |g| dd^c \tau < M_1,$$

where M_1 is a positive constant depending on g .

We now estimate the integration over $A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j)$. If $g(a_j) \neq \infty$. Then $\log^+ |g| \leq C_1$ for some $C_1 > 0$ on $\mathbb{D}(a_j, \frac{3}{2}r_j)$. Then, by (3.7),

$$\int_{A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j)} \log^+ |g| dd^c \tau \leq C_1 \cdot \frac{9\pi}{2} C_h r_j < \frac{1}{2^j},$$

by further shrinking r_j , where $C_h := \max_{\frac{1}{2} \leq s \leq \frac{3}{2}} |h'(s - c)|$. If $g(a_j) = \infty$. Then, locally, $g(z) = (z - a_j)^{k_j} g_j(z)$ with $g_j(a_j) \neq 0$. Thus $\log^+ |g| \leq C_2 \log^+ \frac{1}{|z - a_j|}$ for some $C_2 > 0$ on $\mathbb{D}(a_j, \frac{3}{2}r_j)$. We have similarly

$$\int_{A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j)} \log^+ |g| dd^c \tau \leq \frac{9\pi}{2} C_2 C_h \int_{\frac{1}{2}r_j}^{\frac{3}{2}r_j} \log r dr \leq \frac{9\pi}{2} C_2 C_h r_j^\delta,$$

for some $1 < \delta < 2$. Again, we can choose r_j sufficiently small such that

$$\int_{A(a_j, \frac{1}{2}r_j, \frac{3}{2}r_j)} \log^+ |g| dd^c \tau < \frac{1}{2^j}.$$

Therefore, with this choice of the radii r_j , we obtain from (3.11) that $c_{g,\sigma}^+(r) \leq M_1 + \sum_{j=1}^\infty \frac{1}{2^j} = M_1 + 1$. $\square$

Note that the shrinking process of r_j in the above lemma depends on the given function g . So we must ensure in our arguments that such shrinking occurs only finitely many times. Throughout the rest of the paper, we will keep shrinking the radii r_j whenever necessary. The σ should be understood as being adjusted accordingly, although we will continue to denote it by the same symbol.

3.2. The characteristic functions in the parabolic setting. We continue to let $\mathcal{Y} := \mathbb{C} \setminus \mathcal{E}$ with σ being the exhaustion associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$, constructed in Section 3.1. Let $f : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ be a holomorphic map. We briefly recall some standard notations in Nevanlinna theory in the parabolic context (for details, see [2], [16]).

Let $D = \{Q = 0\}$ be a hypersurface in $\mathbb{P}^n(\mathbb{C})$, where $Q \in \mathbb{C}[x_0, \dots, x_n]$ is a homogeneous polynomial of degree d .

1. The proximity function of f with respect to D is defined as

$$m_{f,\sigma}(D, r) = \int_{S_r^\sigma} \lambda_D(f) d\mu_r,$$

where λ_D is the Weil-function on $\mathbb{P}^n(\mathbb{C})$ given by

$$\lambda_D(\mathbf{x}) = \log \frac{\|\mathbf{x}\|^d}{|Q(\mathbf{x})|}, \quad \mathbf{x} = [x_0 : \cdots : x_n] \in \mathbb{P}^n(\mathbb{C}), \quad \|\mathbf{x}\| := \max\{|x_0|, \dots, |x_n|\}.$$

2. The Nevanlinna characteristic (height) function is defined by

$$T_{f,\sigma}(r) := \int_{S_r^\sigma} \log \|\mathbf{f}\| d\mu_r, \quad r > 1,$$

where $\mathbf{f} = (f_0, \dots, f_n)$ is a reduced representation of f . The Ahlfors-Shimizu characteristic (height) function is defined by

$$T_{f,\sigma}^A(r) = \int_1^r \frac{dt}{t} \int_{B_t^\sigma} f^* \omega_{FS}, \quad r > 1,$$

where ω_{FS} denotes the Fubini-Study form. Note that $T_{f,\sigma}(r)$ depends on the choices of the representation $\mathbf{f}$ of f , while $T_{f,\sigma}^A(r)$ does not depend on the representation. Therefore, when we write $T_{f,\sigma}(r)$, we always assume that we have fixed a reduced representation $\mathbf{f} = (f_0, \dots, f_n)$ of f .

3. The counting function and the k -truncated counting function are defined by

$$N_{f,\sigma}(D, r) := \int_1^r \sum_{z \in B_t^\sigma} \text{ord}_z f^* D \frac{dt}{t}, \quad \text{and} \quad N_{f,\sigma}^{(k)}(D, r) := \int_1^r \sum_{z \in B_t^\sigma} \min\{k, \text{ord}_z f^* D\} \frac{dt}{t}.$$

4. The counting function extends naturally to the sub-schemes, as well as the notion $N_{\text{gcd},\sigma}(f, g)(r)$, where f, g are meromorphic functions on $\mathcal{Y}$. See [13].

We also use the standard notation $T_f(r), m_f(D, r), n_f(D, t)$ and $N_f(D, r)$ of the Nevanlinna functions associated to a holomorphic map $f : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$. In this case, the Ahlfors-Shimizu and the classical Nevanlinna characteristic function agree up to an additive constant (see [14, Theorem 1.1.19]), i.e., $T_f(r) - T_f^A(r) = O(1)$.

Theorem 11 (The parabolic First Main Theorem). *Let $\mathcal{Y} \subset \mathbb{C}$ be as above. Let $D = \{Q = 0\}$ be a hypersurface in $\mathbb{P}^n(\mathbb{C})$, where $Q \in \mathbb{C}[x_0, \dots, x_n]$ is a homogeneous polynomial of degree d . Let $f = (f_0, \dots, f_n) : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ be a holomorphic curve, where $f_0, \dots, f_n$ are entire functions without common zero. Suppose that $f(\mathcal{Y}) \not\subset \text{Supp}(D)$. Then*

$$dT_{f,\sigma}(r) = m_{f,\sigma}(D, r) + N_{f,\sigma}(D, r) + c_{Q(\mathbf{f}),\sigma}(r).$$

Proof. Following the definition of Nevanlinna characteristic function, we have

$$dT_{f,\sigma}(r) = \int_{S_r^\sigma} \log \|\mathbf{f}\|^d d\mu_r = \int_{S_r^\sigma} \log \frac{\|\mathbf{f}\|^d}{|Q(\mathbf{f})|} d\mu_r + \int_{S_r^\sigma} \log |Q(\mathbf{f})| d\mu_r.$$

Hence the assertion follows from Lemma 9 together with the definition of the proximity function. $\square$

In the case of a meromorphic function g on $\mathcal{Y}$, write $g = g_0/g_1$, where g_0 and g_1 are entire functions without common zeros. We regard the function g as the map $g := [g_0 : g_1] : \mathcal{Y} \rightarrow \mathbb{P}^1(\mathbb{C})$. By taking $Q(z_0, z_1) = z_1$ (so $D = \{z_1 = 0\} = \infty$), we get

$$m_{g,\sigma}(r) := m_{g,\sigma}(\infty, r) = \int_{S_r^\sigma} \log \frac{\max\{|g_0|, |g_1|\}}{|g_1|} d\mu_r = \int_{S_r^\sigma} \log^+ |g| d\mu_r + O(1).$$

The above $m_{g,\sigma}(r)$ is well-defined, i.e., it is independent of the choice of g_0 and g_1 . However, the Nevanlinna height function $T_{g,\sigma}(r) = \int_{S_r^\sigma} \log \max\{|g_0|, |g_1|\} d\mu_r$ is not well-defined. Instead we introduce a new height function that is well-defined:

$$(3.12) \quad T_{g,\sigma}^M(r) = m_{g,\sigma}(\infty, r) + N_{g,\sigma}(\infty, r).$$

From the parabolic Jensen's formula above,

$$(3.13) \quad T_{1/g,\sigma}^M(r) = T_{g,\sigma}^M(r) + c_{1/g,\sigma}(r).$$

We now compare $T_{g,\sigma}^M(r)$ with $T_{g,\sigma}^A(r)$. From parabolic Jensen's formula,

$$\begin{aligned} T_{g,\sigma}^A(r) &= \int_1^r \frac{dt}{t} \int_{B_t^\sigma} dd^c \log \max\{|g_0|, |g_1|\} = \int_{S_r^\sigma} \log \max\{|g_0|, |g_1|\} d\mu_r \\ &\quad - \int_{B_r^\sigma} \log \max\{|g_0|, |g_1|\} dd^c \tau \\ &= \int_{S_r^\sigma} \log^+ |g| d\mu_r + \int_{S_r^\sigma} \log |g_1| d\mu_r - \int_{B_r^\sigma} \log \max\{|g_0|, |g_1|\} dd^c \tau \\ &= T_{g,\sigma}^M(r) + \frac{1}{2} \int_{B_r^\sigma} \log(1/(1+|g|^2)) dd^c \tau = T_{g,\sigma}^M(r) + \frac{1}{2} c_{1/(1+|g|^2),\sigma}(r). \end{aligned}$$

Similarly to the proof of Lemma 10, by choosing r_j small enough, there exists a constant $C > 0$ with $c_{1/(1+|g|^2),\sigma}^+(r) \leq C$. Hence

$$(3.14) \quad |T_{g,\sigma}^M(r) - T_{g,\sigma}^A(r)| \leq C.$$

It is important to note that these radii r_j depend on the function g .

Proposition 12 ([2]). *Let $f = (f_0, \dots, f_n) : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ be a holomorphic curve, where $f_0, \dots, f_n$ are entire functions without common zero. Let $\mathcal{Y} \subset \mathbb{C}$ be given with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Then we can choose r_j small enough such that, for all $r \geq 1$, the following hold:*

- (i) $O(1) \leq T_f(r) - T_{f,\sigma}^A(r) \leq \log r + O(1)$.
- (ii) $O(1) \leq T_f(r) - T_{f,\sigma}^M(r) \leq \log r + O(1)$.

Proof. From (3.5), for $t \geq \frac{3}{2}$, $B_t^\sigma \subset \mathbb{D}_t$ and

$$\mathbb{D}_t \setminus \left(\bigcup_{j=1}^{\infty} D(a_j, \frac{3}{2}r_j) \right) \subset B_t^\sigma.$$

Thus

$$\begin{aligned} 0 &\leq T_f^A(r) - T_{f,\sigma}^A(r) \leq \int_{\frac{3}{2}}^r \frac{dt}{t} \int_{(\bigcup_{j=1}^{\infty} \mathbb{D}(a_j, \frac{3}{2}r_j)) \cap \mathbb{D}_t} f^* \omega_{FS} + O(1) \\ &\leq \int_1^r \frac{dt}{t} \int_{\bigcup_{j=1}^{\infty} \mathbb{D}(a_j, \frac{3}{2}r_j)} f^* \omega_{FS} + O(1) \\ &\leq \int_1^r \frac{dt}{t} \left(\sum_{j=1}^{\infty} \int_{\mathbb{D}(a_j, \frac{3}{2}r_j)} f^* \omega_{FS} \right) + O(1), \quad (r \geq 1). \end{aligned}$$

We choose $r_j > 0$ sufficiently small so that

$$(3.15) \quad \int_{\mathbb{D}(a_j, \frac{3}{2}r_j)} f^* \omega_{FS} < 2^{-j}.$$

Hence the above estimate yields

$$0 \leq T_f^A(r) - T_{f,\sigma}^A(r) \leq \log r + O(1), \quad (r \geq 1).$$

By using the fact that $T_f(r) = T_f^A(r) + O(1)$, this completes the proof of (i).

To prove (ii), by using (3.8) with $v := \log \|\mathbf{f}\|$, we get

$$T_{f,\sigma}^A(r) = \int_1^r \frac{dt}{t} \int_{B_t^\sigma} dd^c \log \|\mathbf{f}\| = \int_{S_r^\sigma} \log \|\mathbf{f}\| d\mu_r - \int_{B_r^\sigma} \log \|\mathbf{f}\| dd^c \tau = T_{f,\sigma}(r) - \int_{B_r^\sigma} \log \|\mathbf{f}\| dd^c \tau.$$

Thus

$$|T_{f,\sigma}^A(r) - T_{f,\sigma}(r)| \leq \int_{B_r^\sigma} \log^+ \|\mathbf{f}\| dd^c \tau + \int_{B_r^\sigma} \log^+(1/\|\mathbf{f}\|) dd^c \tau.$$

Since $f_0, \dots, f_n$ have no common zero, we can argue similarly to the proof of Lemma 10 to choose r_j small enough so that there exists a constant $C > 0$ with

$$\int_{B_r^\sigma} \log^+ \|\mathbf{f}\| dd^c \tau + \int_{B_r^\sigma} \log^+ \frac{1}{\|\mathbf{f}\|} dd^c \tau \leq C.$$

for any $r > 0$. By combining above with (i), we prove (ii). $\square$

Again, these radii r_j in Proposition 12 depends on the map f .

3.3. The parabolic Logarithmic Derivative Lemma and H. Cartan's theorem. To establish the parabolic Logarithmic Derivative Lemma, we introduce the weighted Euler characteristic function (see [16, Definition 3.4]):

$$\mathfrak{X}_\sigma^+(r) := \int_{S_r^\sigma} \log^+ \left| d\sigma \left(\frac{\partial}{\partial z} \right) \right|^2 d\mu_r.$$

Using the properties that the disks $\mathbb{D}(a_j, 2r_j)$ are disjoint and that $\sum_{j \geq 1} r_j < +\infty$, we see from [16, Example (2), pp. 32 and 33] (see also [2, (2.10)]) that

$$\mathfrak{X}_\sigma^+(r) = \mathfrak{X}_\sigma(r) + O(\log r) = N_\mathcal{E}(r) + O(\log r), \quad r > 1,$$

where $N_\mathcal{E}(r) := \int_1^r n_\mathcal{E}(t) \frac{dt}{t}$, and $n_\mathcal{E}(t)$ counts the number of points in $\mathcal{E} \cap \mathbb{D}_t$ for $t \geq 1$. In particular, if $\mathcal{E} := f^{-1}(D)$ for a holomorphic map $f : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ and a divisor D on $\mathbb{P}^n(\mathbb{C})$ with $f(\mathbb{C}) \not\subset \text{Supp} D$, it follows (see also [2, (4.2)]) that

$$(3.16) \quad \mathfrak{X}_\sigma^+(r) \leq N_f^{(1)}(D, r) + O(\log r), \quad r > 1.$$

The above estimate is crucial to our paper. In [2], the authors consider the case when $N_f(D, r) = o(T_f(r))$, and therefore $\mathfrak{X}_\sigma^+(r)$ is a small function with respect to $T_f(r)$. In our case, we treat the situation where $N_f^{(1)}(D, r) \leq \frac{1}{\ell} T_f(r)$ for a sufficiently large ℓ . In this setting, $\mathfrak{X}_\sigma^+(r)$ should be regarded as having the same order of growth as $T_f(r)$.

Lemma 13 (Parabolic Logarithmic Derivative Lemma). *Let g be a meromorphic function on $\mathbb{C}$. Let $\mathcal{Y} \subset \mathbb{C}$ be as above. For any $\delta > 0$, one has*

- (i) $m_{g'/g, \sigma}(r) \leq_{\text{exc}} \frac{1}{2}((1 + \delta)^2 \log T_g(r) + \delta \log r + \mathfrak{X}_\sigma^+(r)).$
- (ii) $m_{g^{(l)}/g, \sigma}(r) \leq_{\text{exc}} \frac{l}{2}((1 + \delta)^2 \log T_g(r) + \delta \log r + \mathfrak{X}_\sigma^+(r)).$

Proof. (i) We follow the approach in [16, Theorem 3.8, pp. 26-28]. Consider the auxiliary form

$$\Omega := \frac{1}{|w|^2(1 + \log^2 |w|)} \frac{\sqrt{-1}}{2\pi} dw \wedge d\bar{w},$$

introduced in [16, (15), p. 27], which has finite volume on $\mathbb{C}$. By the change of variable formula, we have

$$\int_1^r \frac{dt}{t} \int_{B_t^\sigma} g^* \Omega = \int_{w \in \mathbb{C}} N_{g, \sigma}(w, r) \Omega.$$

Since $B_t^\sigma \subset \mathbb{D}_t$ for any $t \geq 1$, and by the classical First Main Theorem,

$$N_{g, \sigma}(w, r) \leq N_g(w, r) \leq T_g(r) + O(1), \quad \text{for } r > 1.$$

Thus

$$\int_1^r \frac{dt}{t} \int_{B_t^\sigma} g^* \Omega = \int_{w \in \mathbb{C}} N_{g, \sigma}(w, r) \Omega \leq \int_{\mathbb{C}} T_g(r) \Omega + \int_{\mathbb{C}} C \Omega.$$

Consequently, we obtain

$$\int_1^r \frac{dt}{t} \int_{B_t^\sigma} g^* \Omega \leq C_0 T_g(r)$$

for some constant $C_0 > 0$. The rest of the proof follows from [16, Theorem 3.8, p. 28]. This finishes the proof of (i).

To prove (ii), notice that

$$\frac{g^{(l)}}{g} = \frac{g^{(l)}}{g^{(l-1)}} \frac{g^{(l-1)}}{g^{(l-2)}} \cdots \frac{g'}{g}.$$

Therefore, by (i), we obtain

$$m_{g^{(l)}/g, \sigma}(r) \leq \sum_{j=1}^l m_{g^{(j)}/g^{(j-1)}, \sigma}(r) \leq_{\text{exc}} \frac{1}{2} \left((1+\delta)^2 \sum_{j=1}^l \log T_{g^{(j)}}(r) + \delta l \log r + l \mathfrak{X}_{\sigma}^{+}(r) \right).$$

By the classical First Main Theorem,

$$T_{g^{(j)}}(r) = m_{g^{(j)}}(r) + N_{g^{(j)}}(r) \leq m_{g^{(j)}/g^{(j-1)}}(r) + m_{g^{(j-1)}}(r) + 2N_{g^{(j-1)}}(r) \leq_{\text{exc}} 3T_{g^{(j-1)}}(r) + O(1).$$

Hence

$$T_{g^{(j)}}(r) \leq_{\text{exc}} 3^j T_g(r) + O(1).$$

Therefore, we get

$$m_{g^{(l)}/g, \sigma}(r) \leq_{\text{exc}} \frac{l}{2} \left((1+\delta)^2 \log T_g(r) + \delta \log r + \mathfrak{X}_{\sigma}^{+}(r) \right).$$

□

In particular, by the definition, the above lemma gives the following:

Lemma 14. *If u is an entire function such that it has no zeros on $\mathcal{Y}$ (i.e. u is a unit on $\mathcal{Y}$), then, for any $\delta > 0$,*

$$T_{u'/u, \sigma}^M(r) \leq_{\text{exc}} \frac{1}{2} \left((1+\delta)^2 \log T_u(r) + \delta \log r + \mathfrak{X}_{\sigma}^{+}(r) \right).$$

Using the parabolic Logarithmic Derivative Lemma and the parabolic Jensen's formula, we can easily derive the following parabolic version of H. Cartan's theorem.

Theorem 15. *Let $\mathcal{Y} := \mathbb{C} \setminus \mathcal{E}$ with the exhaustion function σ constructed above. Let $f = (f_0, \dots, f_n) : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ be a holomorphic map whose image is not contained in any proper linear subspace of $\mathbb{P}^n(\mathbb{C})$, where $f_0, \dots, f_n$ are entire functions without common zeros. Let $H_1, \dots, H_q$ be hyperplanes on $\mathbb{P}^n(\mathbb{C})$. Then, for $\delta > 0$,*

$$\begin{aligned} & \int_{S_r^{\sigma}} \max_J \sum_{j \in J} \lambda_{H_j}(f) d\mu_r + N_{W, \sigma}(0, r) \\ & \leq_{\text{exc}} (n+1) T_{f, \sigma}(r) + O(\log T_f(r)) + \frac{n(n+1)}{2} (\mathfrak{X}_{\sigma}^{+}(r) + \delta \log r) + c_{\frac{1}{W}, \sigma}(r), \end{aligned}$$

where $W := W(f_0, \dots, f_n)$ is the Wronskian of f and the maximum is taken over all subsets $J \subset \{1, 2, \dots, q\}$ such that $H_j, j \in J$, are in general position.

Proof. We follow the proof in [17, Theorem A.5.1.3]. Let $H_j = \{\mathbf{w} \in \mathbb{P}^n(\mathbb{C}) \mid \langle \mathbf{w}, \mathbf{a}_j \rangle = 0\}$ for $j = 1, \dots, q$. Let T be the set of all injective maps $\mu : \{0, 1, \dots, n\} \rightarrow \{1, \dots, q\}$ such that $\mathbf{a}_{\mu(0)}, \dots, \mathbf{a}_{\mu(n)}$ are linearly independent. Then

$$\begin{aligned}
 \int_{S_r^\sigma} \max_J \sum_{j \in J} \lambda_{H_j}(f) d\mu_r &\leq \int_{S_r^\sigma} \max_{\mu \in T} \sum_{j=0}^n \log \left(\frac{\|\mathbf{f}\| \|\mathbf{a}_{\mu(j)}\|}{|\langle \mathbf{f}, \mathbf{a}_{\mu(j)} \rangle|} \right) d\mu_r \\
 &\leq \int_{S_r^\sigma} \max_{\mu \in T} \log^+ \frac{|W(\langle \mathbf{f}, \mathbf{a}_{\mu(0)} \rangle, \dots, \langle \mathbf{f}, \mathbf{a}_{\mu(n)} \rangle)|}{\prod_{j=0}^n |\langle \mathbf{f}, \mathbf{a}_{\mu(j)} \rangle|} d\mu_r \\
 (3.17) \quad &+ (n+1)T_{f,\sigma}(r) + \int_{S_r^\sigma} \log(1/|W(f_0, \dots, f_n)|) d\mu_r + O(1).
 \end{aligned}$$

By applying the logarithmic derivative lemma (Lemma 13) and the parabolic Jensen formula (Lemma 9), the theorem follows. $\square$

The following result can be derived from the proof of [19, Theorem 2.1] and Lemma 13.

Theorem 16. *Let $f_0, \dots, f_n, f_{n+1}$ be entire functions without common zeros. Let $\mathcal{Y} := \mathbb{C} \setminus \mathcal{E}$ with the exhaustion function σ constructed above. Assume that $f_0 + \dots + f_n + f_{n+1} = 0$. If $\sum_{i \in I} f_i \neq 0$ for any proper subset $I \subset \{0, \dots, n+1\}$, then we have*

$$T_{f_i/f_j, \sigma}^M(r) \leq_{\text{exc}} \sum_{l=0}^{n+1} N_{f_l, \sigma}^{(n)}(0, r) + \frac{n(n+1)}{2} (\mathfrak{X}_\sigma^+(r) + \log r) + c_{\frac{1}{D_0}, \sigma}(r) + \sum_{\substack{l=0 \\ l \neq j}}^{n+1} c_{f_l, \sigma}(r) + O\left(\max_{1 \leq l \leq n} \log T_{\frac{f_l}{f_0}}(r)\right),$$

where D_0 is an entire function appeared in the proof of [19, Theorem 2.1] or [17, Theorem A5.2.6]), which is determined in terms of $f_0, \dots, f_n$ only.

Proof. We first sketch the proof by assuming that $f_0, \dots, f_n$ are linearly independent. By applying Theorem 15 to $f = (f_0, \dots, f_{n+1})$, we have

$$\sum_{j=1}^{n+2} m_{f, \sigma}(H_j, r) + N_{W, \sigma}(r) \leq (n+1)T_{f, \sigma}(r) + \frac{n(n+1)}{2} (\mathfrak{X}_\sigma^+(r) + \delta \log r) + c_{\frac{1}{W}, \sigma}(r) + O(\log T_f(r)),$$

where $H_i = \{x_{i-1} = 0\}$, $i = 1, \dots, n+1$ and $H_{n+2} = \{x_0 + \dots + x_n = 0\}$. By Theorem 11, for $1 \leq j \leq n+2$,

$$T_{f, \sigma}(r) = m_{f, \sigma}(H_j, r) + N_{f, \sigma}(H_j, r) + c_{f_{j-1}, \sigma}(r).$$

Thus we get,

$$T_{f, \sigma}(r) \leq \sum_{j=0}^{n+1} N_{f_j, \sigma}(0, r) - N_{W, \sigma}(r) + \frac{n(n+1)}{2} (\mathfrak{X}_\sigma^+(r) + \delta \log r) + c_{\frac{1}{W}, \sigma}(r) + \sum_{j=0}^{n+1} c_{f_j, \sigma}(r) + O(\log T_f(r)).$$

It is standard to verify that

$$\sum_{j=0}^{n+1} N_{f_j, \sigma}(0, r) - N_{W, \sigma}(r) \leq \sum_{j=0}^{n+1} N_{f_j, \sigma}^{(n)}(0, r).$$

The proof can be completed by applying the following estimate, valid for any $0 \leq i \leq n$ and $0 \leq j \leq n+1$:

$$\begin{aligned} T_{f, \sigma}(r) &= \int_{S_r^\sigma} \log \max_{0 \leq l \leq n+1} |f_l(z)| d\mu_r = \int_{S_r^\sigma} \log \max_{0 \leq l \leq n+1} (|f_l(z)|/|f_j(z)|) d\mu_r + \int_{S_r^\sigma} \log |f_j(z)| d\mu_r \\ &\geq \int_{S_r^\sigma} \log^+ \left| \frac{f_i(z)}{f_j(z)} \right| d\mu_r + N_{f_j, \sigma}(0, r) + c_{f_j, \sigma}(r) \\ (3.18) \quad &\geq m_{\frac{f_i}{f_j}, \sigma}(r) + N_{\frac{f_i}{f_j}, \sigma}(\infty, r) + c_{f_j, \sigma}(r) = T_{f_i/f_j, \sigma}^M(r) + c_{f_j, \sigma}(r). \end{aligned}$$

Thus

$$T_{f_i/f_j, \sigma}^M(r) \leq \sum_{j=0}^{n+1} N_{f_j, \sigma}^{(n)}(0, r) + \frac{n(n+1)}{2} (\mathfrak{X}_\sigma^+(r) + \delta \log r) + c_{\frac{1}{W}, \sigma}(r) + \sum_{\substack{l=0 \\ l \neq j}}^{n+1} c_{f_l, \sigma}(r) + O\left(\max_{1 \leq l \leq n} \log T_{\frac{f_l}{f_0}}(r)\right).$$

Passing from the condition that $f_0, \dots, f_n$ are linearly independent to the situation in which no proper subsum of $f_0 + \dots + f_{n+1}$ vanishes involves only algebraic arguments. In particular, the resulting statements are obtained from the present ones by changing W to D_0 , the modified Wronskian that appears in the proof of [19, Theorem 2.1]. $\square$

3.4. The moving GCD theorem in the parabolic setting. One of the key components in [9] is the extension of the GCD theorem in [13] to the case where the coefficients of two coprime polynomials are allowed to be meromorphic functions of the same growth. In this subsection, we aim to extend this line of argument to the parabolic setting. Before we proceed, let us briefly outline the development of the GCD theorem.

The foundational arguments in [13] are based on results initially established in [12] for the number field case. In [25], the GCD results concerning $\mathbb{G}_m^n$ from [12] were generalized to more general varieties by combining the approach in [22] with Diophantine approximation results for arbitrary varieties introduced by the first named author and Vojta [21]. More recently, a further generalization was achieved in [10], which applied Vojta's results on integral points on semiabelian varieties for the number field case, as well as [15] for the complex case.

To proceed, we will first derive a version of the second main theorem with moving targets of the same growth, which will serve our purpose. See [18] or [17, Chapter 6].

Let $a_0, \dots, a_n$ be meromorphic functions, not all zeros, on $\mathbb{C}$, and let $L := a_0 X_0 + \dots + a_n X_n$. Then L defines a hyperplane H in $\mathbb{P}^n$, over the field generated by $a_0, \dots, a_n$ over $\mathbb{C}$. We note that

$H(z)$ is the hyperplane determined by the linear form $L(z)$. The definition of the Weil function, proximity function and counting function can be easily extended to moving hyperplanes. For example,

$$(3.19) \quad \lambda_{H(z)}(P) = -\log \frac{|a_0(z)x_0 + \cdots + a_n(z)x_n|}{\max\{|x_0|, \dots, |x_n|\} \max\{|a_0(z)|, \dots, |a_n(z)|\}},$$

for $z \in \mathcal{Y}$ which is not a common zero of $a_0, \dots, a_n$, or a pole of any a_k , $0 \leq k \leq n$. We now consider a family of linear forms $L_j := \sum_{i=0}^n a_{ij}X_j$, $1 \leq j \leq q$, as the aforementioned linear forms. Without loss of generality, we normalize the linear forms L_j , $1 \leq j \leq q$, such that for each $1 \leq j \leq q$, there exists $0 \leq j' \leq n$ with $a_{jj'} = 1$. Let t be a positive integer and let $V(t)$ be the complex vector space spanned by the elements

$$\left\{ \prod a_{jk}^{n_{jk}} : n_{jk} \geq 0, \sum n_{jk} = t \right\},$$

where the products and sums run over $1 \leq j \leq q$ and $0 \leq k \leq n$. Let $1 = b_1, b_2, \dots, b_u$ be a basis of $V(t)$ and $b_1, \dots, b_w$ a basis of $V(t+1)$. It's clear that $u \leq w$. Moreover, we have

$$(3.20) \quad \liminf_{t \rightarrow \infty} \dim V(t+1) / \dim V(t) = 1.$$

See [24, Lemma 6] for a proof.

Definition 17. Let V be a $\mathbb{C}$ -vector space spanned by finitely many meromorphic functions. We say that a holomorphic map $f = (f_0, \dots, f_n) : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ is linearly independent over V if whenever we have a linear combination $\sum_{i=0}^n a_i f_i = 0$ with $a_i \in V$, then $a_i = 0$ for each i . Otherwise we say that f is linearly dependent over V .

The following formulation of the second main theorem with moving targets follows from the strategy in [26] and the proof of [17, Theorem A6.2.4], by adding the Wronskian term and computing the error term explicitly when applying the second main theorem (Theorem 15).

Theorem 18. Let $f = (f_0, \dots, f_n) : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$ be a holomorphic map where $f_0, f_1, \dots, f_n$ are entire functions without common zeros. Let H_j , $1 \leq j \leq q$, be arbitrary (moving) hyperplanes given by $L_j := a_{j0}X_0 + \cdots + a_{jn}X_n$ where $a_{j0}, \dots, a_{jn}$ are meromorphic functions, not all zeros on $\mathbb{C}$. Let t be a positive integer, $1 = b_1, b_2, \dots, b_u$ be a basis of $V(t)$ and $b_1, \dots, b_w$ a basis of $V(t+1)$. Denote by W the Wronskian of $\{hb_m f_k \mid 1 \leq m \leq w, 0 \leq k \leq n\}$, where h is a meromorphic function such that $hb_1, \dots, hb_w$ are entire functions without common zeros. Let $\mathcal{Y} \subset \mathbb{C}$ be as above. If f is linearly non-degenerate over $V(t+1)$, then, we have the following

inequality

$$\begin{aligned}
 & \int_{S_r^\sigma} \max_J \sum_{k \in J} \lambda_{H_k(z)}(f(z)) d\mu_r + \frac{1}{u} N_{W,\sigma}(0, r) \\
 & \leq_{\text{exc}} \left(\frac{w}{u} (n+1) \right) T_{f,\sigma}(r) + \frac{w}{u} (n+1)(t+2) \sum_{j=1}^q T_{L_j,\sigma}(r) \\
 (3.21) \quad & + \frac{w}{2u} (n+1)(nw+w-1)(\mathfrak{X}_\sigma^+(r) + \log r) + \frac{1}{u} c_{\frac{1}{W},\sigma}(r) + O(\log T_f(r)),
 \end{aligned}$$

where $T_{L_j,\sigma}(r) = \sum_{i=0}^n T_{a_{ji},\sigma}^M(r)$, $w = \dim V(t+1)$, $u = \dim V(t)$, and the maximum is taken over all subsets J of $\{1, \dots, q\}$ such that $H_j(z)$, $j \in J$, are in general position.

Remark. Let ϵ be a sufficiently small positive number. By (3.20), we find t such that $\frac{w}{u} \leq 1 + \epsilon$. An explicit upper bound for t can be found in the proof of [3, Proposition 5.3] in term of ϵ and the number of generator for $V(1)$, which is less than qn . More precisely, if $\epsilon < \frac{1}{16}$, then we can find

$$(3.22) \quad t \leq \frac{3qn}{\epsilon} \log \frac{1}{\epsilon}$$

such that $\frac{w}{u} \leq 1 + \epsilon$. Furthermore, for such t , we have

$$(3.23) \quad u := \dim V(t) \leq \left(\frac{1}{\epsilon} \right)^{qn}, \text{ and consequently, } w := \dim V(t+1) \leq (1 + \epsilon) \left(\frac{1}{\epsilon} \right)^{qn}.$$

Using the above theorem, and through the same argument from the proofs of [13, Theorem 5.7], [8, Theorem 26], and [9, Theorem 4.4], we can get the following GCD theorem (note we now consider the case that $u_1, \dots, u_n$ are units on $\mathcal{Y}$). We first fix some notation. Let K be a subfield of the field of meromorphic functions on $\mathbb{C}$. Let F, G be coprime polynomials in n variables of the same degree $d > 0$ over K . Assume that one of the coefficients in the expansions of F and G is equal to 1. For every positive integer t , we denote by $V_{F,G}(t)$ the (finite-dimensional) $\mathbb{C}$ -vector space spanned by $\prod_{\alpha} \alpha^{n_{\alpha}}$, where α runs through all non-zero coefficients of F and G , $n_{\alpha} \geq 0$ and $\sum n_{\alpha} = t$; we also put $d_t := \dim V_{F,G}(t)$. For every integer $m \geq d$, we let $M = M_{m,n,d} = 2 \binom{m+n-d}{n} - \binom{m+n-2d}{n}$. Let $Q = \sum_{\mathbf{i} \in I_Q} a_{\mathbf{i}} \cdot \mathbf{x}^{\mathbf{i}} \in K[x_1, \dots, x_n]$, where $\mathbf{i} = (i_1, \dots, i_n)$, $\mathbf{x}^{\mathbf{i}} = x_1^{i_1} \dots x_n^{i_n}$, and $a_{\mathbf{i}} \neq 0$ if $\mathbf{i} \in I_Q$. We define the parabolic characteristic function of Q as

$$(3.24) \quad T_{Q,\sigma}(r) := \sum_{\mathbf{i} \in I_Q} T_{a_{\mathbf{i}},\sigma}^M(r).$$

Theorem 19 (GCD theorem on parabolic Riemann surfaces evaluating at units). *Let $n \geq 2$ be a positive integer. Let K be a subfield of the field of meromorphic functions on $\mathbb{C}$. Let $\mathcal{Y} \subset \mathbb{C}$ be as above. Let $\mathbf{u} = (u_1, \dots, u_n)$, where $u_1, \dots, u_n$ are meromorphic functions on $\mathbb{C}$ such that each $u_i|_{\mathcal{Y}}$, $1 \leq i \leq n$, is a unit (i.e. without zeros and poles on $\mathcal{Y}$). Let F, G be coprime polynomials*

in n variables of the same degree $d > 0$ over K . Assume that at least one of the coefficients in the expansions of F and G is equal to 1. Let $m \geq d$ and b be a positive integer. If the set $\{u_1^{i_1} \cdots u_n^{i_n} : i_1 + \cdots + i_n \leq m\}$ is linearly independent over $V_{F,G}(Mb+1)$, then we have

$$N_{\gcd,\sigma}(F(\mathbf{u}), G(\mathbf{u}), r) \leq_{\text{exc}} \left(\frac{M'}{M} + \frac{w}{u} - 1 \right) mn \max_{1 \leq i \leq n} T_{u_i, \sigma}^M(r) + O\left(\max_{1 \leq i \leq n} \log T_{u_i}(r) \right) \\ + c'(T_{F,\sigma}(r) + T_{G,\sigma}(r)) + c''(\mathfrak{X}_\sigma^+(r) + \log r + c_{1/h,\sigma}(r) + c_{1/W,\sigma}(r)),$$

where h is meromorphic function such that $F(\mathbf{u})/h$ and $G(\mathbf{u})/h$ are entire and have no common zero, and W is the Wronskian which arises when applying Theorem 18, $w = d_{Mb}$, $u = d_{M(b-1)}$, $c' = \frac{w}{u}M(b+3) + 2$, and $c'' = M\frac{w^2}{u}$, and $M' = \binom{m+n}{n} - M$.

We outline a proof here.

Proof. For a subset $T \subset K[\mathbf{x}]$, where $\mathbf{x} := (x_1, \dots, x_n)$, we denote

$$T_m = \{f \in T : \deg f \leq m\}.$$

As in the proof of [9, Theorem 4.3], we assume that the ideal (F, G) is proper in $K[x]$. Let $(F, G)_m := K[x]_m \cap (F, G)$, where we choose $\{\phi_1, \dots, \phi_M\}$ to be a basis of the K -vector space $(F, G)_m$ consisting of elements of the form $F\mathbf{x}^{\mathbf{i}}, G\mathbf{x}^{\mathbf{i}}$. Let $\Phi := (\phi_1, \dots, \phi_M)$. For each $z \in \mathcal{Y}$, we can construct a basis B_z of $V_m := K[\mathbf{x}]_m / (F, G)_m$ with monomial representatives $\mathbf{x}^{\mathbf{i}_1}, \dots, \mathbf{x}^{\mathbf{i}_{M'}}$ as in the proof of Theorem 5.7 in [13]. Let $I_z := \{\mathbf{i}_1, \dots, \mathbf{i}_{M'}\}$. For each $\mathbf{i} \notin I_z$ with $|\mathbf{i}| = m$, we have

$$\mathbf{x}^{\mathbf{i}} + \sum_{\mathbf{j} \in I_z} c_{z,\mathbf{j}} \mathbf{x}^{\mathbf{j}} \in (F, G)_m$$

for some choice $c_{z,\mathbf{j}} \in K$. Then, for each such $\mathbf{i}$, there is a linear form $\tilde{L}_{z,\mathbf{i}}$ over K such that

$$\tilde{L}_{z,\mathbf{i}}(\phi_1, \dots, \phi_M) = \mathbf{x}^{\mathbf{i}} + \sum_{\mathbf{j} \in I_z} c_{z,\mathbf{j}} \mathbf{x}^{\mathbf{j}}.$$

We note that since there are correspondingly only finitely many choices of a monomial basis of V_m , there are only finitely many choices of $c_{z,\mathbf{j}}$, even as z ranges over all of $\mathcal{Y}$. Following the explicit construction in the proof of [9, Theorem 4.3], we can find a nonzero element $c_z \in V_{F,G}(M)$ such that the normalized linear form $L_{z,\mathbf{i}} := c_z \tilde{L}_{z,\mathbf{i}}$ satisfies

$$L_{z,\mathbf{i}}(\phi_1, \dots, \phi_M) = c_z \left(\mathbf{x}^{\mathbf{i}} + \sum_{\mathbf{j} \in I_z} c_{z,\mathbf{j}} \mathbf{x}^{\mathbf{j}} \right)$$

and

$$(3.25) \quad T_{L_{z,\mathbf{i}},\sigma}(r) \leq M(T_{F,\sigma}(r) + T_{G,\sigma}(r)),$$

by [9, (4.12)]. Moreover, the explicit construction also demonstrates that there are only a finitely many choice for c_z , even though z runs through all of $\mathcal{Y}$.

Following the steps in the proof of Theorem 4.3 in [9], we then obtain

$$(3.26) \quad \log |L_{z,\mathbf{i}}(\Phi(\mathbf{u}))| - \log \|L_{z,\mathbf{i}}\|_z \leq \log |\mathbf{u}(z)^{\mathbf{i}}| + \log \|F\|_z + \log \|G\|_z + O(1).$$

Since the functions u_i are units on $\mathcal{Y}$, and that

$$(3.27) \quad \log^+ |\mathbf{u}(z)^{\mathbf{i}}| \leq \sum_{j=1}^n i_j \log^+ |u_j(z)| \leq m \sum_{j=1}^n \log^+ |u_j(z)|,$$

we have

$$(3.28) \quad T_{\mathbf{u}^{\mathbf{i}},\sigma}^M(r) \leq m \sum_{j=1}^n T_{u_j,\sigma}^M(r) \leq mn \max_{1 \leq i \leq n} T_{u_i,\sigma}^M(r).$$

Consequently, by (3.26), we obtain

$$(3.29) \quad \int_{S_r^\sigma} \sum_{|\mathbf{i}|=m, \mathbf{i} \notin I_z} (\log |L_{z,\mathbf{i}}(\Phi(\mathbf{u}))| - \log \|L_{z,\mathbf{i}}\|_z) d\mu_r \leq M' mn \max_{1 \leq i \leq n} T_{u_i,\sigma}^M(r) + M(T_{F,\sigma}(r) + T_{G,\sigma}(r)) + O(1).$$

Note that the map $\phi_1(\mathbf{u}), \dots, \phi_M(\mathbf{u})$ may not be a reduced presentation of $\Phi(\mathbf{u})$. Let h be a meromorphic function such that $F(\mathbf{u})/h$ and $G(\mathbf{u})/h$ are entire and have no common zero. Let $\psi_i := \phi_i(\mathbf{u})/h$ for $i = 1, \dots, M$. Then $\Psi := (\psi_1, \dots, \psi_M)$ is a reduced representation of $\Phi(\mathbf{u})$. Then

$$(3.30) \quad \int_{S_r^\sigma} \log \|\Phi(\mathbf{u})\| d\mu_r = \int_{S_r^\sigma} \log \|\Psi\| d\mu_r + \int_{S_r^\sigma} \log |h| d\mu_r = T_{\Psi,\sigma}(r) + \int_{S_r^\sigma} \log |h| d\mu_r.$$

On the other hand, since u_i are units on $\mathcal{Y}$, the poles of h only arises from the poles of coefficients of F and G . More precisely,

$$(3.31) \quad N_{h,\sigma}(\infty, r) \leq T_{F,\sigma}(r) + T_{G,\sigma}(r)$$

by applying the inequality of the first main theorem in Theorem 11. It is also clear that this choice of h gives

$$(3.32) \quad N_{h,\sigma}(0, r) = N_{\gcd,\sigma}(F(\mathbf{u}), G(\mathbf{u}), r).$$

By Lemma 9, (3.31) and (3.32), we have

$$(3.33) \quad \begin{aligned} \int_{S_r^\sigma} \log |h| d\mu_r &= N_{h,\sigma}(0, r) - N_{h,\sigma}(\infty, r) + c_{h,\sigma}(r) \\ &\geq N_{\gcd,\sigma}(F(\mathbf{u}), G(\mathbf{u}), r) - (T_{F,\sigma}(r) + T_{G,\sigma}(r)) + c_{h,\sigma}(r). \end{aligned}$$

Then we can derive from (3.30) that

$$(3.34) \quad \int_{S_r^\sigma} \log \|\Phi(\mathbf{u})\| d\mu_r \geq T_{\Psi,\sigma}(r) + N_{\gcd,\sigma}(F(\mathbf{u}), G(\mathbf{u}), r) + c_{h,\sigma}(r) - (T_{F,\sigma}(r) + T_{G,\sigma}(r)).$$

Since the set $\{u_1^{i_1} \cdots u_n^{i_n} : i_1 + \cdots + i_n \leq m\}$ is linearly independent over $V_{F,G}(Mb + 1)$, it follows that $\Phi(\mathbf{u}) : \mathbb{C} \rightarrow \mathbb{P}^{M-1}$ is linearly independent over $V_{F,G}(Mb) = V(b)$. We note that, when computing Weil functions, we have the freedom to assume that the coefficients of $L_{z,\mathbf{i}}$ are

entire functions without common zeros, by multiplying a suitable meromorphic function. By this construction, for each $z \in \mathcal{Y}$, the hyperplanes defined by evaluating the linear forms $L_{z,\mathbf{i}}$ at z with $|\mathbf{i}| = m$ and $\mathbf{i} \notin I_z$ are in general position. For the computation (3.25) and (3.29), however, we will use the original $L_{z,\mathbf{i}}$. By applying Theorem 18 to $\Phi(\mathbf{u})$ and the set of the linear forms $\{L_{z,\mathbf{i}} : |\mathbf{i}| = m, \mathbf{i} \notin I_z\}$, we have the following inequality:

$$\begin{aligned}
 & \int_{S_r^\sigma} \sum_{|\mathbf{i}|=m, \mathbf{i} \notin I_z} \lambda_{L_{z,\mathbf{i}}}(\Phi(\mathbf{u})) d\mu_r + \frac{1}{u} N_{W,\sigma}(0, r) \\
 & \leq_{\text{exc}} \left(\frac{w}{u} M + \epsilon \right) T_{\Psi,\sigma}(r) + \frac{w}{u} M^2 (b+2) (T_{F,\sigma}(r) + T_{G,\sigma}(r)) \\
 (3.35) \quad & + M^2 \frac{w^2}{u} (\mathfrak{X}_\sigma^+(r) + \log r) + \frac{1}{u} c_{1/W,\sigma}(r) + O(\log \max_{1 \leq j \leq n} T_{u_j}(r)),
 \end{aligned}$$

where

$$\lambda_{L_{z,\mathbf{i}}}(\Phi(\mathbf{u})) = -\log |L_{z,\mathbf{i}}(\Phi(\mathbf{u}))| + \log \|\Phi(\mathbf{u})\| + \log \|L_{z,\mathbf{i}}\| + O(1),$$

and W is the wronskian of $\psi_1, \dots, \psi_M$. By combining (3.35) with (3.29), (3.34) and use (3.33), we get

$$\begin{aligned}
 & MN_{\text{gcd},\sigma}(F(\mathbf{u}), G(\mathbf{u}), r) + \frac{1}{u} N_{W,\sigma}(0, r) \\
 & \leq_{\text{exc}} \left(\frac{w}{u} M - M + \epsilon \right) T_{\Psi,\sigma}(r) + M' mn \max_{1 \leq i \leq n} T_{u_i,\sigma}^M(r) \\
 & + M \left(\frac{w}{u} M(b+2) + 1 \right) (T_{F,\sigma}(r) + T_{G,\sigma}(r)) \\
 & + M^2 \frac{w^2}{u} (\mathfrak{X}_\sigma^+(r) + \log r) + \frac{1}{u} c_{\frac{1}{W},\sigma}(r) + M c_{\frac{1}{h},\sigma}(r) + O(1).
 \end{aligned}$$

Finally, we need to estimate $T_{\Psi,\sigma}(r)$ to complete the proof. Recall that $\psi_i = \phi(\mathbf{u})/h$ and ϕ_i is of the form $F\mathbf{x}^{\mathbf{i}}$, or $G\mathbf{x}^{\mathbf{i}}$, which is a combination of monomial of degree $\leq m$ with coefficients coming from F or G . Therefore, by (3.27)

$$\log \|\psi_i\| \leq m \sum_{j=1}^n \log^+ |u_j| + \log \|F\| + \log \|G\| + \log \frac{1}{h}.$$

Combining this with Lemma 9 and (3.31), we obtain

$$T_{\Psi,\sigma}(r) \leq mn \max_{1 \leq i \leq n} T_{u_i,\sigma}^M(r) + 2(T_{F,\sigma}(r) + T_{G,\sigma}(r)) + c_{\frac{1}{h},\sigma}(r).$$

□

Theorem 20 (The key theorem). *Let $\mathcal{Y} \subset \mathbb{C}$ be given as above with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Let $u_1, \dots, u_n$, with $n \geq 2$, be meromorphic functions on $\mathbb{C}$ such that they are units on $\mathcal{Y}$. Let K be a subfield of the field of meromorphic functions on $\mathbb{C}$. Let F and G be coprime polynomials in $K[x_1, \dots, x_n]$. Assume that at least one of the coefficients in the expansions of F and G is equal to 1. Let $\epsilon > 0$. Then, by shrinking r_j small enough (depending*

on ϵ , u_i , $1 \leq i \leq n$, and on F and G), we can choose a positive integer m and positive reals c_i , $0 \leq i \leq 5$, all depending only on ϵ , n , and the degree of F and G such that, either

$$(3.36) \quad T_{u_1^{m_1} \dots u_n^{m_n}, \sigma}^M(r) \leq_{\text{exc}} c_1 \max_{1 \leq i \leq n} \{\log T_{u_i}(r)\} + c_2(T_{F, \sigma}(r) + T_{G, \sigma}(r)) + c_3(\mathfrak{X}_\sigma^+(r) + \log r)$$

for some non-trivial tuple of integers $(m_1, \dots, m_n)$ with $|m_1| + \dots + |m_n| \leq 2m$, or

$$(3.37) \quad \begin{aligned} N_{\text{gcd}, \sigma}(F(u_1, \dots, u_n), G(u_1, \dots, u_n), r) &\leq_{\text{exc}} \epsilon \max_{1 \leq i \leq n} \{T_{u_i}(r)\} + c_4(T_{F, \sigma}(r) + T_{G, \sigma}(r)) \\ &\quad + c_5(\mathfrak{X}_\sigma^+(r) + \log r). \end{aligned}$$

Furthermore, if $\epsilon > 0$ is sufficiently small, then we may choose $m = C_1 \epsilon^{-1}$, where C_1 is a positive real independent of ϵ .

Proof. We note that c_i , $1 \leq i \leq 5$, will be determined during the course of the proof. Recall that $M := M_m := 2 \binom{m+n-d}{n} - \binom{m+n-2d}{n}$ and $M' := M'_m := \binom{m+n}{n} - M = O(m^{n-2})$. Based on the above estimates, we can choose a real $C_1 \geq 1$ independent of ϵ and the u_i , $1 \leq i \leq n$, such that $m = C_1 \epsilon^{-1} \geq 2d$, and

$$(3.38) \quad \frac{M' m n}{M} \leq \frac{\epsilon}{4}.$$

By (3.20) we may then choose a sufficiently large integer $b \in \mathbb{N}$ such that

$$(3.39) \quad \frac{w}{u} - 1 \leq \frac{\epsilon}{4mn},$$

where $w := \dim_{\mathbf{k}} V_{F, G}(Mb)$ and $u := \dim_{\mathbf{k}} V_{F, G}(Mb - M)$.

Suppose that the set $\{u_1^{i_1} \dots u_n^{i_n} : i_1 + \dots + i_n \leq m\}$ is linearly independent over $V_{F, G}(Mb + 1)$. Then, Theorem 19 implies that, by noticing (3.38) and (3.39),

$$\begin{aligned} N_{\text{gcd}, \sigma}(F(\mathbf{u}), G(\mathbf{u}), r) &\leq_{\text{exc}} \epsilon \max_{1 \leq i \leq n} \{T_{u_i, \sigma}^M(r)\} + \tilde{c} \max_{1 \leq i \leq n} \{\log T_{u_i}(r)\} \\ &\quad + c_4(T_{F, \sigma}(r) + T_{G, \sigma}(r)) + c_5 \left(\mathfrak{X}_\sigma^+(r) + \log r + c_{\frac{1}{h}, \sigma}(r) + c_{\frac{1}{W}, \sigma}(r) \right), \end{aligned}$$

where $\tilde{c}$ and c_i , $4 \leq i \leq 5$, are computable positive reals.

If the set $\{u_1^{i_1} \dots u_n^{i_n} : i_1 + \dots + i_n \leq m\}$ is linearly dependent over $V_{F, G}(Mb + 1)$, i.e. there is a non-trivial relation

$$(3.40) \quad \sum_{\mathbf{i} \in I} \alpha_{\mathbf{i}} \mathbf{u}^{\mathbf{i}} = 0,$$

where the sum runs over those $\mathbf{i} \in \mathbb{Z}_{\geq 0}^n$ and $|\mathbf{i}| \leq m$, $\alpha_{\mathbf{i}} \in V_{F, G}(Mb + 1)$ for each $\mathbf{i}$, and $\alpha_{\mathbf{i}} \neq 0$ for $\mathbf{i} \in I$. Without loss of generality, we assume that no proper subsum of (3.40) vanishes. Then

Theorem 16 implies that

$$(3.41) \quad \begin{aligned} T_{\frac{\alpha_{\mathbf{i}}}{\alpha_{\mathbf{i}_0}} \mathbf{u}^{\mathbf{i}-\mathbf{i}_0}, \sigma}(r) &\leq_{\text{exc}} \sum_{\mathbf{i} \in I} N_{\beta \alpha_{\mathbf{i}}, \sigma}(0, r) + c_3(\mathfrak{X}_{\sigma}^+(r) + \log r) \\ &+ c_{\frac{1}{D_0}, \sigma}(r) + \sum_{\substack{\mathbf{i} \in I \\ \mathbf{i} \neq \mathbf{i}_0}} c_{\beta \alpha_{\mathbf{i}} \mathbf{u}^{\mathbf{i}}, \sigma}(r) + O\left(\max_{0 \leq i \leq n} \{\log T_{u_i}(r)\}\right) + O(\log(T_F(r) + \log F_G(r))), \end{aligned}$$

where $c_3 = \binom{n+m}{n} \left(\binom{n+m}{n} + 1 \right)$. We note that

$$(3.42) \quad N_{\beta \alpha_{\mathbf{i}}, \sigma}(0, r) \leq N_{\alpha_{\mathbf{i}}, \sigma}(0, r) + N_{\beta, \sigma}(0, r) \leq 2(Mb + 1)(T_{F, \sigma}(r) + T_{G, \sigma}(r)) + c_{\frac{1}{\alpha_{\mathbf{i}}}, \sigma}(r),$$

since the zero of β comes from the pole of $\alpha_{\mathbf{i}}$ and by (3.13). Also, since $\alpha_{\mathbf{i}} \in V_{F, G}(Mb + 1)$,

$$(3.43) \quad \begin{aligned} T_{\frac{\alpha_{\mathbf{i}_0}}{\alpha_{\mathbf{i}}}, \sigma}(r) &\leq T_{1/\alpha_{\mathbf{i}}, \sigma}^M(r) + T_{\alpha_{\mathbf{i}_0}, \sigma}^M(r) \leq T_{\alpha_{\mathbf{i}}, \sigma}^M(r) + T_{\alpha_{\mathbf{i}_0}, \sigma}^M(r) + c_{\frac{1}{\alpha_{\mathbf{i}}}, \sigma}(r) \\ &\leq 2(Mb + 1)(T_{F, \sigma}(r) + T_{G, \sigma}(r)) + c_{\frac{1}{\alpha_{\mathbf{i}}}, \sigma}(r). \end{aligned}$$

Then, by combining (3.41), (3.42), (3.43),

$$\begin{aligned} T_{\mathbf{u}^{\mathbf{i}-\mathbf{i}_0}, \sigma}(r) &\leq T_{\frac{\alpha_{\mathbf{i}}}{\alpha_{\mathbf{i}_0}} \mathbf{u}^{\mathbf{i}-\mathbf{i}_0}, \sigma}(r) + T_{\frac{\alpha_{\mathbf{i}_0}}{\alpha_{\mathbf{i}}}, \sigma}(r) \leq_{\text{exc}} c_2(T_{F, \sigma}(r) + T_{G, \sigma}(r)) + c_3(\mathfrak{X}_{\sigma}^+(r) + \log r) \\ &+ c_{\frac{1}{D_0}, \sigma}(r) + 2c_{\frac{1}{\alpha_{\mathbf{i}}}, \sigma}(r) + \sum_{\substack{\mathbf{i} \in I \\ \mathbf{i} \neq \mathbf{i}_0}} c_{\beta \alpha_{\mathbf{i}} \mathbf{u}^{\mathbf{i}}, \sigma}(r) + O\left(\max_{0 \leq i \leq n} \{\log T_{u_i}(r)\}\right), \end{aligned}$$

where $c_2 = 4(Mb + 1)$.

By Lemma 10, we can shrink r_j small enough so that all of the quantities

$$c_{1/h, \sigma}^+(r), \quad c_{1/W, \sigma}^+(r), \quad c_{1/D_0, \sigma}(r), \quad c_{1/\alpha_{\mathbf{i}}, \sigma}(r) \quad (\mathbf{i} \in I), \quad \sum_{\substack{\mathbf{i} \in I \\ \mathbf{i} \neq \mathbf{i}_0}} c_{\beta \alpha_{\mathbf{i}} \mathbf{u}^{\mathbf{i}}, \sigma}(r)$$

are bounded by $O(1)$, where the implied constant depends on $\mathbf{u}$, F , and G , but is independent of r . In addition, we shrink r_j small enough such that (3.14) and the Proposition 12 for u_i , $1 \leq i \leq n$, hold, i.e. $T_{u_i}(r)$ and $T_{u_i, \sigma}^M(r)$ differ only by $\log r + O(1)$. This proves the key theorem. $\square$

For the convenience of later application, we state the following result for $n = 1$, which can be easily obtained from the theory of resultant (see [20, Proposition 10]).

Proposition 21. *Let K be a subfield of the field of meromorphic functions on $\mathbb{C}$. Let $F, G \in K[x]$ be nonconstant coprime polynomials. Let $\mathcal{Y} \subset \mathbb{C}$ be given as above with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Then there exists a positive constant c such that for any meromorphic function g on $\mathcal{Y}$, the following holds:*

$$N_{\text{gcd}, \sigma}(F(g), G(g), r) \leq c(T_{F, \sigma}(r) + T_{G, \sigma}(r)).$$

4. THE ABC THEOREM FOR PARABOLIC RIEMANN SURFACES

Let $\mathcal{Y} \subset \mathbb{C}$ be given as above with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Let $\mathbf{u} = (u_1, \dots, u_n)$ where $u_1, \dots, u_n$ are meromorphic functions on $\mathbb{C}$ such that each restriction $u_i|_{\mathcal{Y}}$, $1 \leq i \leq n$, is a unit. Let $\mathbb{K}_{\mathbf{u}}$ be the set of meromorphic function a on $\mathbb{C}$ for which there exists a nonnegative constant c_a (depending on a) with

$$(4.1) \quad T_{a,\sigma}^M(r) \leq_{\text{exc}} c_a \left(\mathfrak{X}_{\sigma}^+(r) + \log r + \max_{1 \leq i \leq n} \{\log T_{u_i}(r)\} \right),$$

after shrinking radii r_j sufficiently small (depending on a). By (3.13) and Lemma 10, it is easy to see that $\mathbb{K}_{\mathbf{u}}$ is a subfield of the meromorphic function field. Notably, in our application, the constant c_a can be effectively computed. We will use Theorem 20 for $K := \mathbb{K}_{\mathbf{u}}$. Note that, by Lemma 13 and by (3.13), if $a \in \mathbb{K}_{\mathbf{u}}$, then $a' \in \mathbb{K}_{\mathbf{u}}$, since

$$\begin{aligned} T_{a',\sigma}^M(r) &\leq T_{\frac{a'}{a},\sigma}^M(r) + T_{a,\sigma}^M(r) \leq m_{a'/a,\sigma}(r) + N_{a,\sigma}(0, r) + N_{a,\sigma}(\infty, r) + T_{a,\sigma}^M(r) \\ &\leq m_{a'/a,\sigma}(r) + T_{\frac{1}{a},\sigma}^M(r) + 2T_{a,\sigma}^M(r). \end{aligned}$$

Let $\mathbf{x} := (x_1, \dots, x_n)$. For $\mathbf{i} = (i_1, \dots, i_n) \in \mathbb{Z}^n$, we let $\mathbf{x}^{\mathbf{i}} := x_1^{i_1} \cdots x_n^{i_n}$ and $\mathbf{u}^{\mathbf{i}} := u_1^{i_1} \cdots u_n^{i_n}$. For a non-constant polynomial $F(\mathbf{x}) = \sum_{\mathbf{i}} a_{\mathbf{i}} \mathbf{x}^{\mathbf{i}} \in \mathbb{K}_{\mathbf{u}}[\mathbf{x}] := \mathbb{K}_{\mathbf{u}}[x_1, \dots, x_n]$, we define

$$(4.2) \quad D_{\mathbf{u}}(F)(\mathbf{x}) := \sum_{\mathbf{i}} \frac{(a_{\mathbf{i}} \mathbf{u}^{\mathbf{i}})'}{\mathbf{u}^{\mathbf{i}}} \mathbf{x}^{\mathbf{i}} = \sum_{\mathbf{i}} \left(a'_{\mathbf{i}} + a_{\mathbf{i}} \cdot \sum_{j=1}^n i_j \frac{u'_j}{u_j} \right) \mathbf{x}^{\mathbf{i}} \in \mathbb{K}_{\mathbf{u}}[\mathbf{x}].$$

A direct computation shows that

$$(4.3) \quad F(\mathbf{u})' = D_{\mathbf{u}}(F)(\mathbf{u}),$$

and that the following product rule

$$(4.4) \quad D_{\mathbf{u}}(FG) = D_{\mathbf{u}}(F)G + FD_{\mathbf{u}}(G)$$

holds for $F, G \in \mathbb{K}_{\mathbf{u}}[\mathbf{x}]$.

By Lemma 14, we see that $u'_i/u_i \in \mathbb{K}_{\mathbf{u}}$. In the following lemma and in Theorem 23, we will identify certain monomial products of bounded degree in the u_i that satisfy (4.1), and hence are contained in $\mathbb{K}_{\mathbf{u}}$.

The following lemma shares the same proof of Lemma 5.1 in [9].

Lemma 22. *Let $\mathbf{u} = (u_1, \dots, u_n)$, where u_i , $1 \leq i \leq n$, are the units on $\mathcal{Y}$. Let G be a non-constant polynomial in $\mathbb{K}_{\mathbf{u}}[\mathbf{x}]$ with no monomial factors and no repeated factors, and assume that at least one of its coefficients is equal to 1. Then $F := D_{\mathbf{u}}(G)$ and G are coprime in $\mathbb{K}_{\mathbf{u}}[\mathbf{x}]$ unless there exists a non-trivial tuple of integers $(m_1, \dots, m_n)$ with $\sum_{i=1}^n |m_i| \leq 2 \deg G$ such that $T_{u_1^{m_1} \cdots u_n^{m_n}, \sigma}^M \leq T_{G, \sigma}(r)$.*

Proof. Relation (4.4) implies that we may assume, as in the first step of the proof of [9, Lemma 5.1], that G is irreducible. Write $G = \sum_{\mathbf{i}} b_{\mathbf{i}} \mathbf{x}^{\mathbf{i}} \in \mathbb{K}_{\mathbf{u}}[\mathbf{x}]$, which contains at least two distinct terms, since G has no monomial factors. Then $D_{\mathbf{u}}(G)/G \in \mathbb{C}^*$ since $D_{\mathbf{u}}(G)$ is not zero and $\deg D_{\mathbf{u}}(G) = \deg G$. Comparing the coefficients of G and $D_{\mathbf{u}}(G)$, for nonzero $b_{\mathbf{i}}$ and $b_{\mathbf{j}} = 1$ in $\mathbb{K}_{\mathbf{u}}$ ($\mathbf{i} \neq \mathbf{j}$), we have from (4.2) that

$$\frac{(b_{\mathbf{i}} \mathbf{u}^{\mathbf{i}})'}{b_{\mathbf{i}} \mathbf{u}^{\mathbf{i}}} = \frac{(\mathbf{u}^{\mathbf{j}})'}{\mathbf{u}^{\mathbf{j}}},$$

where $\mathbf{i} = (i_1, \dots, i_n)$, and $\mathbf{j} = (j_1, \dots, j_n)$. It implies that $b_{\mathbf{i}} \frac{\mathbf{u}^{\mathbf{i}}}{\mathbf{u}^{\mathbf{j}}} \in \mathbb{C}$. Therefore,

$$T_{u_1^{j_1-i_1} \dots u_n^{j_n-i_n}, \sigma}^M \leq T_{G, \sigma}(r).$$

□

As an application of the key theorem (Theorem 20) and using the above lemma, we obtain the following theorem.

Theorem 23. *Let $\mathcal{Y} \subset \mathbb{C}$ be given as above with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Let $\mathbf{u} = (u_1, \dots, u_n)$, where each u_i , $1 \leq i \leq n$, is a meromorphic function on $\mathbb{C}$ which is a unit on $\mathcal{Y}$. Let G be a non-constant polynomial in $\mathbb{K}_{\mathbf{u}}[\mathbf{x}]$ with no monomial factors and no repeated factors. Assume that at least one of the coefficients in the expansions of G is equal to 1. Then, by shrinking r_j small enough (depending on ϵ , $\mathbf{u}$ and G), we can find a positive integer m and positive constants $\tilde{c}_1$ and $\tilde{c}_2$ depending only on ϵ , n , and on the degree and the coefficients of G , but independent of $\mathbf{u}$, for which the following holds: Either there exists a non-trivial n -tuple $(m_1, \dots, m_n)$ of integers with $\sum_{i=1}^n |m_i| \leq 2m$ such that*

$$(4.5) \quad T_{u_1^{m_1} \dots u_n^{m_n}, \sigma}^M(r) \leq_{\text{exc}} \tilde{c}_1 (\mathfrak{X}_{\sigma}^+(r) + \log r) + O\left(\max_{1 \leq j \leq n} \{\log T_{u_j}(r)\}\right),$$

or else

$$(4.6) \quad N_{G(\mathbf{u}), \sigma}(0, r) - N_{G(\mathbf{u}), \sigma}^{(1)}(0, r) \leq_{\text{exc}} \epsilon \max_{1 \leq j \leq n} \{T_{u_j}(r)\} + \tilde{c}_2 (\mathfrak{X}_{\sigma}^+(r) + \log r).$$

Proof. We first note that the constant $\tilde{c}_1, \tilde{c}_2$ will be determined during the course of the proof. Let $z_0 \in \mathcal{Y}$. If $v_{z_0}(G(\mathbf{u})) \geq 2$, then it follows from (4.3) that $v_{z_0}(D_{\mathbf{u}}(G)(\mathbf{u})) = v_{z_0}(G(\mathbf{u})) - 1$. Hence,

$$\min\{v_{z_0}^+(G(\mathbf{u})), v_{z_0}^+(D_{\mathbf{u}}(G)(\mathbf{u}))\} \geq v_{z_0}^+(G(\mathbf{u})) - \min\{1, v_{z_0}^+(G(\mathbf{u}))\}.$$

Consequently,

$$(4.7) \quad N_{\text{gcd}, \sigma}(G(\mathbf{u}), D_{\mathbf{u}}(G)(\mathbf{u}), r) \geq N_{G(\mathbf{u}), \sigma}(0, r) - N_{G(\mathbf{u}), \sigma}^{(1)}(0, r).$$

Write $G = \sum_{\mathbf{i}} a_{\mathbf{i}} \mathbf{x}^{\mathbf{i}}$ with $a_{\mathbf{i}} \in \mathbb{K}_{\mathbf{u}}$. Then, from the definition of $\mathbb{K}_{\mathbf{u}}$, there exists a nonnegative real c_G (depending on G) such that

$$(4.8) \quad T_{G,\sigma}(r) \leq_{\text{exc}} c_G \left(\mathfrak{X}_{\sigma}^+(r) + \log r + \max_{1 \leq i \leq n} \{\log T_{u_i}(r)\} \right)$$

after shrinking radii r_j sufficiently small (depending on G). Furthermore, by Lemma 13, we have

$$(4.9) \quad \begin{aligned} T_{D_{\mathbf{u}}(G),\sigma}(r) &\leq 2T_{G,\sigma}(r) + \deg G \sum_{i=1}^n T_{u'_i/u_i,\sigma}^M(r) \\ &\leq_{\text{exc}} (2c_G + n \deg G) (\mathfrak{X}_{\sigma}^+(r) + \log r) + O\left(\max_{1 \leq i \leq n} \{\log T_{u_i}(r)\}\right). \end{aligned}$$

By Lemma 22 and (4.8), either G and $D_{\mathbf{u}}(G)$ are coprime or there exists a non-trivial n -tuple $(m_1, \dots, m_n) \in \mathbb{Z}^n$ with $\sum_{i=1}^n |m_i| \leq 2 \deg G$ such that

$$T_{u_1^{m_1} \dots u_n^{m_n}, \sigma}^M(r) \leq_{\text{exc}} T_{G,\sigma}(r) \leq_{\text{exc}} c_G (\mathfrak{X}_{\sigma}^+(r) + \log r) + O\left(\max_{1 \leq i \leq n} \{\log T_{u_i}(r)\}\right).$$

It remains to consider where G and $D_{\mathbf{u}}(G)$ are coprime. Let $\epsilon > 0$ be given, which may assume to be sufficiently small. By apply Theorem 20, after shrinking r_j small enough (depending on ϵ , $\mathbf{u}$ and G), we find m and $c_i > 0$, $1 \leq i \leq 4$, such that either

$$(4.10) \quad T_{u_1^{m_1} \dots u_n^{m_n}, \sigma}^M(r) \leq_{\text{exc}} c_1 (T_{G,\sigma}(r) + T_{D_{\mathbf{u}}(G),\sigma}(r)) + c_2 (\mathfrak{X}_{\sigma}^+(r) + \log r)$$

for some non-trivial tuple of integers $(m_1, \dots, m_n)$ with $|m_1| + \dots + |m_n| \leq 2m$, or

$$(4.11) \quad \begin{aligned} &N_{\text{gcd},\sigma}(G(\mathbf{u}), D_{\mathbf{u}}(G)(\mathbf{u}), r) \\ &\leq_{\text{exc}} \frac{\epsilon}{2} \max_{1 \leq i \leq n} T_{u_i}(r) + c_3 (T_{G,\sigma}(r) + T_{D_{\mathbf{u}}(G),\sigma}(r)) + c_4 (\mathfrak{X}_{\sigma}^+(r) + \log r). \end{aligned}$$

We can conclude the proof by combining (4.8), (4.9), (4.10), and (4.11). $\square$

The proof of [7, Theorem 4] can be adapted to the current situation, leading to the following result.

Theorem 24. *Let G be a non-constant polynomial in $\mathbb{C}[x_1, \dots, x_n]$ with no monomial factors and no repeated factors. Then, for any $\epsilon > 0$, there exist a positive real number c_0 , a proper Zariski closed subset Z of $\mathbb{A}^n(\mathbb{C})$ expressible as the zero locus of a finite set $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$ with the following property: Let $\mathbf{u} = (u_1, \dots, u_n)$, where each u_i , $1 \leq i \leq n$, is meromorphic function on $\mathbb{C}$, with $H(\mathbf{u}) \neq 0$ for all $H \in \Sigma$ and $G(\mathbf{u}) \neq 0$. Let $\mathcal{Y} \subset \mathbb{C}$ be given as above with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Assume that $u_1, \dots, u_n$ are units on $\mathcal{Y}$. Then, by shrinking r_j small enough (depending on $\mathbf{u}$ and G), we have*

$$(4.12) \quad N_{G(\mathbf{u}),\sigma}(0, r) - N_{G(\mathbf{u}),\sigma}^{(1)}(0, r) \leq_{\text{exc}} \epsilon \max_{1 \leq i \leq n} \{\log T_{u_i}(r)\} + c_0 (\mathfrak{X}_{\sigma}^+(r) + \log r).$$

Here, c_0 can be effectively bounded from above in terms of ϵ , n , and the degree of G . Moreover, the finite set $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$ satisfies: (Z1) Σ depends on ϵ and G , but not on $\mathbf{u}$ or $\mathcal{Y}$, and

can be determined explicitly, (Z2) $|\Sigma|$ and the degree of each polynomial in Σ can be effectively bounded from above in terms of ϵ , n , and the degree of G .

We outline a proof here.

Proof. We follow the proof of [7, Theorem 4]. We apply Theorem 23 through several reduction steps to derive a contradiction. The constant c_0 will be determined in the course of the proof. We also remark that, since the coefficients of G lie in $\mathbb{C}$, the assumption that at least one coefficient in the expansion of G is equal to 1 imposes no restriction. By Theorem 23, after shrinking r_j , we have either

$$(4.13) \quad T_{u_1^{m_1} \dots u_n^{m_n}, \sigma}^M(r) \leq_{\text{exc}} \tilde{c}_1(\mathfrak{X}_\sigma^+(r) + \log r) + O(\max_{1 \leq j \leq n} \{\log T_{u_j}(r)\})$$

or

$$(4.14) \quad N_{G(\mathbf{u}), \sigma}(0, r) - N_{G(\mathbf{u}), \sigma}^{(1)}(0, r) \leq_{\text{exc}} \epsilon \max_{1 \leq j \leq n} \{T_{u_j}(r)\} + \tilde{c}_2(\mathfrak{X}_\sigma^+(r) + \log r).$$

Let $c_0 \geq \tilde{c}_2$. If (4.14) holds, then we are done. Assume it is not, then (4.13) holds for a non-trivial n -tuple $(m_1, \dots, m_n)$ of integers with $\sum |m_i| \leq M_1$, where M_1 , depending only on ϵ , n , and the degree of G , but independent of $\mathbf{u}$. Thus, we have

$$(4.15) \quad \lambda_1 := u_1^{m_1} \dots u_n^{m_n} \in \mathbb{K}_{\mathbf{u}}.$$

We may assume $\gcd(m_1, \dots, m_n) = 1$. By Lemma 14 in [7], $(m_1, \dots, m_n)$ extends to a basis $(m_1, \dots, m_n), (a_{21}, \dots, a_{2n}), \dots, (a_{n1}, \dots, a_{nn})$ of $\mathbb{Z}^n$ with

$$(4.16) \quad |a_{i1}| + \dots + |a_{in}| \leq M_1 + n \quad \text{for } 2 \leq i \leq n.$$

Consider the change of variables

$$(4.17) \quad \Lambda_1 := x_1^{m_1} \dots x_n^{m_n}, \quad \text{and} \quad X_{1,i} := x_1^{a_{i1}} \dots x_n^{a_{in}} \quad \text{for } 2 \leq i \leq n$$

and put

$$(4.18) \quad \beta_{1,i} = u_1^{a_{i1}} \dots u_n^{a_{in}} \quad \text{for } 2 \leq i \leq n.$$

Let A_1 denote the $n \times n$ matrix whose rows are the above basis of $\mathbb{Z}^n$. Then we formally express the above identities as

$$(4.19) \quad (\Lambda_1, X_{1,2}, \dots, X_{1,n}) = (x_1, \dots, x_n)^{A_1} \quad \text{and} \quad (\lambda_1, \beta_{1,2}, \dots, \beta_{1,n}) = (u_1, \dots, u_n)^{A_1}.$$

Let $B_1 = A_1^{-1}$. The entries of B_1 can be bounded from above in terms of M_1 and n . We have

$$(4.20) \quad (x_1, \dots, x_n) = (\Lambda_1, X_{1,2}, \dots, X_{1,n})^{B_1} \quad \text{and} \quad (u_1, \dots, u_n) = (\lambda_1, \beta_{1,2}, \dots, \beta_{1,n})^{B_1}.$$

Let $G_1(\Lambda_1, X_{1,2}, \dots, X_{1,n}) \in \mathbb{C}[\Lambda_1, X_{1,2}, \dots, X_{1,n}]$ with no monomial factors and

$$(4.21) \quad G((\Lambda_1, X_{1,2}, \dots, X_{1,n})^{B_1}) = \Lambda_1^{d_1} X_{1,2}^{d_2} \dots X_{1,n}^{d_n} G_1(\Lambda_1, X_{1,2}, \dots, X_{1,n})$$

for some integers d_i , $1 \leq i \leq n$. Since the transformations in (4.19) and (4.20) are inverses of each other, and G has no repeated irreducible factors, it follows that G_1 also has no repeated irreducible factors. The coefficients of G_1 are the same as the coefficients of G and $\deg(G_1)$ can be bounded from above explicitly in terms of M_1 , n , and $\deg(G)$. Consider $G_1(\lambda_1, X_{1,2}, \dots, X_{1,n}) \in \mathbb{C}(\lambda_1)[X_{1,2}, \dots, X_{1,n}] \subset \mathbb{K}_{\mathbf{u}}[X_{1,2}, \dots, X_{1,n}]$.

For the particular change of variables in (4.19), (4.20), and (4.21) (that depends on the matrix A_1), we apply the Lemma 16 in [7] to find a nonconstant polynomial $\tilde{H}_1 \in \mathbb{C}[x_1, \dots, x_n]$ such that $G_1(\lambda_1, X_{1,2}, \dots, X_{1,n})$ has neither monomial nor repeated irreducible factors if $\tilde{H}_1(u_1, \dots, u_n) \neq 0$. It is important to note that the polynomial $\tilde{H}_1$ is not only independent of $\mathbf{u}$, but also independent of $\mathcal{Y}$. We now take H_1 to be the product of all such $\tilde{H}_1$ where A_1 ranges over the finitely many elements of $\text{GL}_n(\mathbb{Z})$ with $\|A_1\|_\infty \leq M_1 + n$. From Lemma 16 in [7] we know that $\deg H_1$ depends only on ϵ , n and $\deg G$.

Since the u_i 's, λ_1 , and $\beta_{1,j}$'s are the units on $\mathcal{Y}$, we have

$$(4.22) \quad N_{G(\mathbf{u}),\sigma}(0, r) - N_{G(\mathbf{u}),\sigma}^{(1)}(0, r) = N_{G_1(\lambda_1, \beta_{1,2}, \dots, \beta_{1,n}),\sigma}(0, r) - N_{G_1(\lambda_1, \beta_{1,2}, \dots, \beta_{1,n}),\sigma}^{(1)}(0, r).$$

By (4.19) and (4.20), we have:

$$(4.23) \quad \max_{1 \leq i \leq n} \{T_{u_i}(r)\} = O\left(\max\{T_{\lambda_1}(r), T_{\beta_{1,2}}(r), \dots, T_{\beta_{1,n}}(r)\}\right).$$

By (4.16), there are only a finitely many choice of $\beta_{1,t}$, $2 \leq t \leq n$. Hence, by Proposition 12, after possibly shrinking r_j (depending on u_i), we may assume that

$$|T_{u_i}(r) - T_{u_i,\sigma}^M(r)|, \quad |T_{\beta_{1,t}}(r) - T_{\beta_{1,t},\sigma}^M(r)|, \quad \text{and} \quad |T_{\lambda_1}(r) - T_{\lambda_1,\sigma}^M(r)|$$

are all bounded by $\log r + O(1)$ for $1 \leq i \leq n$ and $2 \leq t \leq n$. Then by (4.23) and (4.13), we find positive real numbers $b_{1,1}$ and $b_{1,2}$ such that

$$b_{1,1} \max_{1 \leq i \leq n} \{T_{u_i}(r)\} \leq_{\text{exc}} \max_{2 \leq t \leq n} \{T_{\beta_{1,t}}(r)\} + \tilde{c}_1(\mathfrak{X}_\sigma^+(r) + \log r), \quad \text{and}$$

$$\max_{2 \leq t \leq n} \{T_{\beta_{1,t}}(r)\} \leq b_{1,2} \left(\max_{1 \leq i \leq n} \{T_{u_i}(r)\} + \log r \right).$$

In conclusion, at the end of this step we have that, after possibly shrinking r_j , the inequality

$$(4.24) \quad N_{G_1(\lambda_1, \beta_{1,2}, \dots, \beta_{1,n}),\sigma}(0, r) - N_{G_1(\lambda_1, \beta_{1,2}, \dots, \beta_{1,n}),\sigma}^{(1)}(0, r) \leq_{\text{exc}} \frac{\epsilon}{b_{1,2}} \max_{2 \leq i \leq n} \{T_{\beta_{1,i}}(r)\} + c_0(\mathfrak{X}_\sigma^+(r) + \log r)$$

fails to hold under the assumption that $H_1(u_1, \dots, u_n) \neq 0$.

The above is the first step. The next step is to apply Theorem 23 to G_1 and $\mathbf{u}$. Note that, to do so, we need to further shrink r_j according to G_1 . There are $(n-1)$ many steps in total as in the proof of [7, Theorem 4]. Let $2 \leq s \leq n-1$ and suppose that we have completed the step $s-1$. This includes the construction of $H_{s-1} \in \mathbb{C}[x_1, \dots, x_n]$ with degree depends on ϵ , n

and $\deg G$ only. The last one is Step $(n-1)$ resulting in $H_{n-1} \in \mathbb{C}[x_1, \dots, x_n]$. We now define $H = H_1 \cdots H_{n-1}$. Then $\deg H$ depends only on ϵ , n and $\deg G$, since each H_i does so. Moreover, the polynomial H is not only independent of $\mathbf{u}$, but also independent of $\mathcal{Y}$.

Suppose $H(u_1, \dots, u_n) \neq 0$. Assume we go through all the $(n-1)$ steps to get the polynomial

$$P(X_{n-1,n}) := G_{n-1}(\lambda_1, \dots, \lambda_{n-1}, X_{n-1,n}) \in \mathbb{K}_{\mathbf{u}}[X_{n-1,n}]$$

such that its degree can be bounded explicitly in terms of $M_{n-1}, \dots, M_1$, n , and $\deg(G)$. At the end of the step $(n-1)$, we have that $\beta := \beta_{n-1,n}$, which is a (non-constant) unit on $\mathcal{Y}$, such that the inequality, after further shrinking r_j according to $G, G_1, \dots, G_{n-1}$,

$$(4.25) \quad N_{P(\beta),\sigma}(0, r) - N_{P(\beta),\sigma}^{(1)}(0, r) \leq_{\text{exc}} \frac{\epsilon}{b_{n-1,2}} T_{\beta}(r) + c_0(\mathfrak{X}_{\sigma}^+(r) + \log r)$$

fails to hold. Here, $b_{n-1,1}$ and $b_{n-1,2}$ are computable positive real numbers. On the other hand, since $H_{n-1}(u_1, \dots, u_n) \neq 0$, the polynomial $P(X_{n-1,n})$ has neither monomial nor repeated irreducible factors, and thus, similar to (4.7), we have

$$(4.26) \quad N_{\text{gcd},\sigma}(P(\beta), D_{\beta}(P)(\beta), r) \geq N_{P(\beta),\sigma}(0, r) - N_{P(\beta),\sigma}^{(1)}(0, r).$$

By Proposition 21, there is a positive constant C_1 independent of β such that

$$(4.27) \quad N_{\text{gcd},\sigma}(P(\beta), D_{\beta}(P)(\beta), r) \leq C_1(T_{P,\sigma}(r) + T_{D_{\beta}(P),\sigma}(r)).$$

Similar to the arguments for (4.9), after further shrinking r_j according to P , we find a positive constant C_2 independent of β such that

$$(4.28) \quad T_{P,\sigma}(r) + T_{D_{\beta}(P),\sigma}(r) \leq_{\text{exc}} C_2(\mathfrak{X}_{\sigma}^+(r) + \log r) + O(\log T_{\beta}(r)).$$

Hence,

$$(4.29) \quad N_{P(\beta),\sigma}(0, r) - N_{P(\beta),\sigma}^{(1)}(0, r) \leq_{\text{exc}} C_1 C_2(\mathfrak{X}_{\sigma}^+(r) + \log r) + O(\log T_{\beta}(r)),$$

which contradicts that fact that (4.25) fails to hold by taking $c_0 \geq C_1 C_2$. $\square$

Theorem 25. *Let $\mathcal{Y} \subset \mathbb{C}$ be given as above with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Let $G = \sum_{\mathbf{i} \in I_G} \alpha_{\mathbf{i}} \mathbf{x}^{\mathbf{i}}$ be a non-constant polynomial in $\mathbb{C}[x_1, \dots, x_n]$. Assume that $G(0, \dots, 0) \neq 0$ and $\deg_{x_i} G = \deg G = d$ for $1 \leq i \leq n$. Let $f_0, \dots, f_n$ be entire functions without common zero which are units on $\mathcal{Y}$. Let $u_i := f_i/f_0$, $1 \leq i \leq n$, and $\mathbf{u} = (u_1, \dots, u_n)$. Assume that $\sum_{\mathbf{i} \in J} \alpha_{\mathbf{i}} \mathbf{u}^{\mathbf{i}} \neq 0$ for any non-empty subset J of I_G . Then, for every $\epsilon > 0$, by shrinking r_j small enough (depending on ϵ , $\mathbf{u}$ and G), we have*

$$(4.30) \quad N_{G(\mathbf{u}),\sigma}(0, r) \geq_{\text{exc}} (\deg G - \epsilon) \cdot \max_{1 \leq i \leq n} \{T_{u_i}(r)\} - c(\mathfrak{X}_{\sigma}^+(r) + \log r),$$

where $c = \frac{1}{2} \binom{n+d}{n} (\binom{n+d}{n} + 1)$.

Proof. Since $G(0, \dots, 0) \neq 0$ and $\deg_{x_i} G = \deg G = d$ for $1 \leq i \leq n$, we may assume without loss of generality that $G(0, \dots, 0) = 1$, and write

$$(4.31) \quad G(\mathbf{u}) = 1 + \sum_{1 \leq i \leq n} \alpha_i u_i^d + \sum_{\mathbf{i} \in I_G \setminus I} \alpha_i \mathbf{u}^{\mathbf{i}}$$

where $\alpha_i \neq 0$ for $0 \leq i \leq n$ and $I = \{\mathbf{i}_0 := (0, 0, \dots, 0), \mathbf{i}_1 := (d, 0, \dots, 0), \dots, \mathbf{i}_n := (0, \dots, 0, d)\}$. Since $\sum_{\mathbf{i} \in J} \alpha_i \mathbf{u}^{\mathbf{i}} \neq 0$, for any non-empty subset J of I_G , we may use Theorem 16 to show that

$$d \max_{1 \leq i \leq n} \{T_{u_i, \sigma}^M(r)\} \leq N_{G(\mathbf{u}), \sigma}(0, r) + c(\mathfrak{X}_{\sigma}^+(r) + \log r) + c_{h_{\mathbf{u}}, \sigma}^+(r) + O(\max_{1 \leq i \leq n} \log T_{u_i}(r)),$$

where $c = \frac{1}{2} \binom{n+d}{n} (\binom{n+d}{n} + 1)$, and $h_{\mathbf{u}}$ is a meromorphic function depending on $\mathbf{u}$. By Lemma 10, by shrinking r_j small enough if necessary, we can make $c_{h_{\mathbf{u}}, \sigma}^+(r) \leq O(1)$. In addition, we can shrink r_j small enough such that (3.14) and the Proposition 12 for u_i , $1 \leq i \leq n$, hold, i.e., $|T_{u_i, \sigma}^M(r) - T_{u_i}(r)| \leq \log r + O(1)$. This proves the theorem. $\square$

Theorem 26 (The abc Theorem on parabolic Riemann surfaces). *Let $\mathcal{Y} \subset \mathbb{C}$ be given with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$. Let G be a non-constant polynomial in $\mathbb{C}[x_1, \dots, x_n]$ with no monomial factors and no repeated factors. Then, for any $\epsilon > 0$, there exist a positive real number c_0 , a proper Zariski closed subset W of $\mathbb{A}^n(\mathbb{C})$ expressible as the zero locus of a finite set $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$ with the following property: Let $f_0, \dots, f_n$ be entire functions without common zeros, which are units on $\mathcal{Y}$. For $1 \leq i \leq n$, set $u_i := f_i/f_0$ and write $\mathbf{u} = (u_1, \dots, u_n)$. Assume that $H(\mathbf{u}) \neq 0$ for all $H \in \Sigma$. Then, by shrinking r_j small enough (depending on ϵ , $\mathbf{u}$ and G), we have*

- (a) $N_{G(\mathbf{u}), \sigma}(0, r) - N_{G(\mathbf{u}), \sigma}^{(1)}(0, r) \leq_{\text{exc}} \epsilon \max_{1 \leq i \leq n} \{T_{u_i}(r)\} + c_0(\mathfrak{X}_{\sigma}^+(r) + \log r)$.
- (b) If $G(0, \dots, 0) \neq 0$ and $\deg_{x_i} G = \deg G = d$ for $1 \leq i \leq n$, then

$$(\deg G - \epsilon) \cdot \max_{1 \leq i \leq n} \{T_{u_i}(r)\} \leq_{\text{exc}} N_{G(\mathbf{u}), \sigma}^{(1)}(0, r) + c_1(\mathfrak{X}_{\sigma}^+(r) + \log r).$$

Here, c_0 and c_1 can be effectively bounded from above in terms of ϵ , n , and the degree of G . Moreover, the finite set $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$ satisfies: (Z1) Σ depends on ϵ and G , but not on $\mathbf{u}$ or $\mathcal{Y}$, and can be determined explicitly, (Z2) $|\Sigma|$ and the degree of each polynomial in Σ can be effectively bounded from above in terms of ϵ , n , and the degree of G .

Proof. The proof follows from Theorem 24 and Theorem 25. Let $W = Z \cup Z_1 \cup [G = 0]$, where Z and Z_1 are the exceptional sets arising from Theorem 24 and Theorem 25 for $\frac{\epsilon}{2}$, respectively. We express the zero locus of W as the vanishing set of a finite collection $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$, and assume that $H(\mathbf{u}) \neq 0$ for all $H \in \Sigma$. By Theorem 24, we obtain

$$N_{G(\mathbf{u}), \sigma}(0, r) - N_{G(\mathbf{u}), \sigma}^{(1)}(0, r) \leq \frac{\epsilon}{2} \max_{1 \leq i \leq n} \{T_{u_i}(r)\} + c_0(\mathfrak{X}_{\sigma}^+(r) + \log r).$$

On the other hand, from (4.30) in Theorem 25,

$$N_{G(\mathbf{u}),\sigma}(0, r) \geq_{\text{exc}} (\deg G - \epsilon) \cdot \max_{1 \leq i \leq n} \{T_{u_i}(r)\} - c(\mathfrak{X}_\sigma^+(r) + \log r).$$

Thus we derive

$$(4.32) \quad N_{G(\mathbf{u}),\sigma}^{(1)}(0, r) \geq_{\text{exc}} \left(\deg G - \frac{\epsilon}{2} \right) \max_{1 \leq i \leq n} \{T_{u_i}(r)\} - c(\mathfrak{X}_\sigma^+(r) + \log r).$$

Let $c_1 = c_0 + c$, then

$$N_{G(\mathbf{u}),\sigma}^{(1)}(0, r) \geq_{\text{exc}} (\deg G - \epsilon) \cdot \max_{1 \leq i \leq n} \{T_{u_i}(r)\} + c_1(\mathfrak{X}_\sigma^+(r) + \log r).$$

□

5. PROOF OF THE MAIN THEOREM

Proof of the Main Theorem. Let $f = (f_0, \dots, f_n) : \mathbb{C} \mapsto \mathbb{P}^n(\mathbb{C})$, where $f_0, \dots, f_n$ are entire functions without common zero. We may assume f_i is not identically zero by letting Z contain all coordinate hyperplanes. When f is rational, the assertion follows directly from Theorem 6, as in this case we have

$$N_{G(f)}(0, r) = N(G(f)) \log r, \quad N_{G(f)}^{(1)}(0, r) = N^{(1)}(G(f)) \log r, \quad \text{and } T_f(r) = (\deg f) \log r$$

for r sufficiently large. Therefore, it remains to consider the situation when f is not rational. In this case, we have

$$\log r = o(T_f(r)).$$

Let $D := \sum_{i=0}^n H_i$, where $H_i := [x_i = 0]$, $0 \leq i \leq n$, are the coordinate hyperplanes. We consider $\mathcal{Y}_f := \mathbb{C} \setminus f^{-1}(D)$ with σ being associated to the discs $\{\mathbb{D}(a_j, 2r_j)\}$, as being constructed in Section 3.1. Furthermore, by further shrinking r_j , we may assume that

$$(5.1) \quad |T_{u_i}(r) - T_{u_i,\sigma}^M(r)| \leq \log r + O(1)$$

for $1 \leq i \leq n$ by (3.14) and Proposition 12 (ii), and

$$(5.2) \quad \mathfrak{X}_\sigma^+(r) \leq \sum_{i=0}^n N_f^{(1)}(H_i, r) + O(\log r)$$

by (3.16). Since $\text{mult}_{z_0}(f^*H_i) \geq \ell$ for all i with $0 \leq i \leq n$ and for every $z_0 \in \mathbb{C}$, we have $N_f^{(1)}(H_i, r) \leq \frac{1}{\ell} T_f(r)$ for $i = 0, 1, \dots, n$. Thus

$$(5.3) \quad \mathfrak{X}_\sigma^+(r) \leq \frac{n+1}{\ell} T_f(r) + O(\log r),$$

where ℓ is a positive integer to be determined.

Let $u_i := \frac{f_i}{f_0}$. Then $u_i, 1 \leq i \leq n$, are units on $\mathcal{Y}_f$. Let $\tilde{G} \in \mathbb{C}[x_1, \dots, x_n]$ be defined by $\tilde{G}(x_1, \dots, x_n) = G(1, x_1, \dots, x_n)$. Then $f_0^{\deg G} \cdot \tilde{G}(\mathbf{u}) = G(f)$. Hence it is obvious that

$$(5.4) \quad N_{\tilde{G}(\mathbf{u}), \sigma}^{(1)}(0, r) = N_{G(f), \sigma}^{(1)}(0, r).$$

Let $\epsilon > 0$, and let W be the proper Zariski closed subset W of $\mathbb{A}^n$ determined in Theorem 26. We express the zero locus of W as the vanishing set of a finite collection $\Sigma \subset \mathbb{C}[x_1, \dots, x_n]$, and assume that $H(\mathbf{u}) \neq 0$ for all $H \in \Sigma$. Then, by Theorem 26, together with (5.1) and (5.4), we obtain, after shrinking r_j if necessary, that there exists a positive real c_0 (depending only on ϵ , n , and $\deg G$) such that

$$(5.5) \quad N_{G(f), \sigma}(0, r) - N_{G(f), \sigma}^{(1)}(0, r) \leq_{\text{exc}} \frac{\epsilon}{3} \max_{1 \leq i \leq n} \{T_{u_i}(r)\} + c_0(\mathfrak{X}_{\sigma}^+(r) + \log r),$$

and

$$(5.6) \quad \left(\deg G - \frac{\epsilon}{2}\right) \cdot \max_{1 \leq i \leq n} \{T_{u_i}(r)\} \leq_{\text{exc}} N_{G(f), \sigma}^{(1)}(0, r) + c_0(\mathfrak{X}_{\sigma}^+(r) + \log r).$$

Using (5.3) and the fact that $\log r = o(T_f(r))$, by taking $\ell > 3c_0(n+2)\epsilon^{-1}$, we have that

$$(5.7) \quad c_0(\mathfrak{X}_{\sigma}^+(r) + \log r) \leq \epsilon/3 \max_{1 \leq i \leq n} \{T_{u_i}(r)\}.$$

Since $B_t^{\sigma} \subset \mathbb{D}_t$ for any $t \geq 1$, $N_{G(f), \sigma}^{(1)}(0, r) \leq N_{G(f)}^{(1)}(0, r)$. It follows immediately from (5.6) and (5.7) that

$$(5.8) \quad N_{G(f)}^{(1)}(0, r) \geq_{\text{exc}} \left(\deg G - \frac{5\epsilon}{6}\right) \cdot \max_{1 \leq i \leq n} \{T_{u_i}(r)\} \geq (\deg G - \epsilon)T_f(r).$$

This proves (ii).

It remains to prove (i), which follows from (ii), i.e., from (5.8) together with the First Main Theorem.

$$N_{G(f)}(0, r) - N_{G(f)}^{(1)}(0, r) \leq_{\text{exc}} \deg G \cdot T_f(r) - (\deg G - \epsilon)T_f(r) \leq_{\text{exc}} \epsilon T_f(r).$$

Finally, we can conclude the proof by taking Z to be the Zariski closure of W in $\mathbb{P}^n(\mathbb{C})$. $\square$

6. TORIC VARIETIES

Proof of Theorem 3. We recall the following setup of finding a natural finite open covering of X from the proof of [12, Theorem 4.4]. Let Σ be the fan corresponding to the smooth projective toric variety X . Then there is a finite affine covering $\{X_{\eta}\}$ of X , where $\eta \in \Sigma$ is a n -dimensional smooth cone with an isomorphism $i_{\eta} : X_{\eta} \rightarrow \mathbb{A}^n$. This isomorphism restricts to an automorphism of $\mathbb{G}_m^n$, where we identify $\mathbb{G}_m^n \subset X_{\eta}$ naturally as a subset of X and $\mathbb{G}_m^n \subset \mathbb{A}^n$ in the standard way such that $\mathbb{A}^n \setminus \mathbb{G}_m^n$ consists of the affine coordinate hyperplanes $\{x_i = 0\}$, $1 \leq i \leq n$. Under this identification, an entire orbifold curve f into (X, Δ) corresponds to a collection of n -tuples

of meromorphic functions $i_\eta(f) = (u_{\eta,1}, \dots, u_{\eta,n})$ such that the zero multiplicity of each $u_{\eta,i}$, $1 \leq i \leq n$, is at least ℓ if not zero. If $z_0 \in \mathbb{C}$ is a pole of some $u_{\eta,i}$, $1 \leq i \leq n$, then $f(z_0)$ is in a component of D_0 , the boundary of X , with multiplicity

$$-\min\{0, v_{z_0}(u_{\eta,1}), \dots, v_{z_0}(u_{\eta,n})\} \geq \ell.$$

Let $\tilde{i}_\eta : X_\eta \rightarrow \mathbb{P}^n(\mathbb{C})$ be the composition of i_η with the natural embedding of $\mathbb{A}^n$ to $\mathbb{P}^n(\mathbb{C})$. Then $\tilde{i}_\eta(f) = (f_{\eta,0}, \dots, f_{\eta,n}) = [1 : u_{\eta,1} : \dots : u_{\eta,n}] : \mathbb{C} \rightarrow \mathbb{P}^n(\mathbb{C})$, where $f_{\eta,0}, \dots, f_{\eta,n}$ are entire functions with no common zero. Furthermore, the pullback $(\tilde{i}_\eta^{-1})^*(D|_{X_\eta})$ of D to $\mathbb{P}^n(\mathbb{C})$ is defined by some nonconstant homogeneous polynomial $F_\eta \in \mathbb{C}[x_0, \dots, x_n]$ with no monomial and repeated factors and $\deg_{X_i} F_\eta = \deg F_\eta$ for each $0 \leq i \leq n$ since D is reduced and is in general position with the boundary of $\mathbb{G}_m^n$ in X . By [27, Proposition 12.11], there exist a non-zero constant $c_{\eta,A}$ and a proper closed subset $Z_{\eta,A} \subset X_\eta$, depending on η and A such that

$$(6.1) \quad T_{\tilde{i}_\eta(f)}(r) \leq c_{\eta,A} T_{A,f}(r) + O(1)$$

if the image of $i_\eta(f)$ is not contained in $Z_{\eta,A}$. Let M be the number of X_η in Σ . We may now apply the Main Theorem to each X_η . Let $\epsilon > 0$. Then there exists a proper Zariski closed subset $W_\eta \subset X_\eta$ for each $\eta \in \Sigma$ such that for sufficiently large integer ℓ , we have

$$(6.2) \quad N_{F_\eta(\tilde{i}_\eta(f))}(0, r) - N_{F_\eta(\tilde{i}_\eta(f))}^{(1)}(0, r) \leq_{\text{exc}} \frac{\epsilon}{M c_{\eta,A}} T_{\tilde{i}_\eta(f)}(r),$$

and

$$(6.3) \quad N_{F_\eta(\tilde{i}_\eta(f))}^{(1)}(0, r) \geq_{\text{exc}} \left(\deg F_\eta - \frac{\epsilon}{M c_{\eta,A}} \right) T_{\tilde{i}_\eta(f)}(r),$$

for all f such that the image of f restricting to X_η is not contained in W_η . Since $\deg_{X_i} F_\eta = \deg F_\eta$ for each $0 \leq i \leq n$, it implies that $\deg F_\eta \cdot T_{\tilde{i}_\eta(f)}(r) = T_{F_\eta(\tilde{i}_\eta(f))}(r) + O(1)$. Then by the first main theorem and (6.3), we have

$$(6.4) \quad m_{F_\eta(\tilde{i}_\eta(f))}(0, r) \leq_{\text{exc}} \frac{\epsilon}{M c_{\eta,A}} T_{\tilde{i}_\eta(f)}(r).$$

Let W be the union of W_η and $Z_{\eta,A}$ for $\eta \in \Sigma$. Then

$$(6.5) \quad N_f(D, r) - N_f^{(1)}(D, r) \leq \sum_{\eta \in \Sigma} N_{F_\eta(\tilde{i}_\eta(f))}(0, r) - N_{F_\eta(\tilde{i}_\eta(f))}^{(1)}(0, r) \leq_{\text{exc}} \epsilon T_{A,f}(r),$$

and

$$(6.6) \quad m_f(D, r) \leq \sum_{\eta \in \Sigma} m_{F_\eta(\tilde{i}_\eta(f))}(0, r) \leq_{\text{exc}} \epsilon T_{A,f}(r)$$

for all orbifold curve $f : \mathbb{C} \rightarrow (X, \Delta)$ with multiplicity at least ℓ along D_0 and $f(\mathbb{C}) \not\subset W$. Finally, the assertion (ii) follows from (6.5) and the following inequality:

$$N_f(D, r) = T_{D,f}(r) - m_f(D, r) \geq_{\text{exc}} T_{D,f}(r) - \epsilon T_{A,f}(r)$$

by (6.6). □

Proof of Theorem 4. Let $\pi : Y \rightarrow X$ be a finite morphism. We recall from [4, Lemma 1] and the proof of [6, Theorem 6] that R is a big divisor since (Y, Δ) is of general type. Let R_0 be an irreducible component of R . Without loss of generality, we let $R_0 = R$. Otherwise, we simply repeat the following steps for each irreducible component. Let $D = \pi(R)$, which is in general position with D_0 by assumption. Now, consider $f : \mathbb{C} \rightarrow (Y, \Delta)$ as a nonconstant orbifold entire curve. Since π^*D has multiplicity at least 2 along R , we have

$$(6.7) \quad N_f(R, r) \leq N_{\pi \circ f}(D, r) - N_{\pi \circ f}^{(1)}(D, r).$$

We are now in a position to apply Theorem 3 for a small $\epsilon > 0$ (to be determined later) and an ample divisor A on X . Consequently, there exists a positive integer ℓ_1 and a proper Zariski closed subset W_1 of X such that if the orbifold (X, Δ_0) has multiplicities at least ℓ_1 along D_0 , then

$$(6.8) \quad N_{\pi \circ f}(D, r) - N_{\pi \circ f}^{(1)}(D, r) \leq_{\text{exc}} \epsilon T_{A, \pi \circ f}(r),$$

and

$$(6.9) \quad T_{D, \pi \circ f}(r) \leq_{\text{exc}} N_{\pi \circ f}^{(1)}(D, r) + \epsilon T_{A, \pi \circ f}(r)$$

if the image of $\pi \circ f$ is not contained in W_1 . Assuming that $\ell \geq \ell_1$ and the image of f is not contained in $\pi^{-1}(W_1)$, then by (6.7) and (6.8), we have

$$(6.10) \quad N_f(R, r) \leq_{\text{exc}} \epsilon T_{A, \pi \circ f}(r).$$

Since $R \leq \pi^*(D)$ (as divisors), the functorial property implies that

$$(6.11) \quad m_f(R, r) \leq m_{\pi \circ f}(D, r) + O(1) = T_{D, \pi \circ f}(r) - N_{\pi \circ f}(D, r) + O(1)$$

Together with (6.9), we have:

$$(6.12) \quad m_f(R, r) \leq \epsilon T_{A, \pi \circ f}(r) + O(1).$$

Combining the two equations (6.10) and (6.12) and using the functorial property of characteristic functions, we have:

$$(6.13) \quad T_{R, f}(r) \leq 2\epsilon T_{\pi^*A, f}(r) + O(1).$$

Since R is a big divisor on Y , by [27, Proposition 12.11], there exists a constant c and a proper Zariski-closed subset W_0 of Y , depending only on A and R , such that

$$(6.14) \quad T_{\pi^*A, f}(r) \leq c T_{R, f}(r) + O(1),$$

if the image of f is not contained in W_0 . Combining (6.14) with (6.13), we obtain

$$T_{R, f}(r) \leq_{\text{exc}} 2c\epsilon T_{R, f}(r) + O(1).$$

This is not possible as c is independent of ϵ , and ϵ can be taken sufficiently small. In conclusion, we have found a sufficiently large integer ℓ such that if (Y, Δ) has multiplicities at least ℓ along D_0 , then the image of any orbifold entire curve f on (Y, Δ) is contained in $\pi^{-1}(W_1) \cup W_0$. $\square$

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REFERENCES

- [1] F. CAMPANA, *Orbifolds, special varieties and classification theory*, Ann. Inst. Fourier (Grenoble) **54** (2004), no. 3, 499–630.
- [2] Z. CHEN, D.T. HUYNH, R. SUN AND S.Y. XIE, *Entire holomorphic curves into $\mathbb{P}^n(\mathbb{C})$ intersecting $n + 1$ general hypersurfaces*, Science China Mathematics, **68**(2025), 2847–2866.
- [3] P. CORVAJA, J. DEMEIO, D. MASSER AND U. ZANNIER, *On the torsion values for sections of an elliptic scheme*, J. Reine Angew. Math., **782**(2022), 1–41.
- [4] P. CORVAJA AND U. ZANNIER, *Algebraic hyperbolicity of ramified covers of $\mathbb{G}_m^2$ (and integral points on affine subsets of $\mathbb{P}^2$)*, J. Differential Geom. **93** (2013), no. 3, 355–377.
- [5] D. A. COX, J. B. LITTLE, AND H. K. SCHENCK, *Toric varieties*, Graduate Studies in Mathematics, 124. American Mathematical Society, Providence, RI, 2011.
- [6] C. GASBARRI, J. GUO AND J. T.-Y. WANG, *Campana conjecture for coverings of toric varieties over function fields*, Algebraic Geometry, to appear, arXiv:2401.13186.
- [7] J. GUO, K. D. NGUYEN, C.-L. SUN AND J. T.-Y. WANG, *Vojta's abc Conjecture for algebraic tori and applications over function fields*, Advances in Mathematics, Paper No. 110358, 37 pp., 2025
- [8] J. GUO, C.-L. SUN AND J. T.-Y. WANG, *On Pisot's d -th root conjecture for function fields and related GCD estimates*, J. of Number Theory, **231**(2022), 401–432.
- [9] J. GUO AND J. T.-Y. WANG, *A complex case of Vojta's general abc conjecture and cases of Campana's orbifold conjecture*, Trans. Amer. Math. Soc. **377**(2024), no. 7, 4956–4991.
- [10] K. HUANG AND A. LEVIN, *Greatest Common Divisors on the Complement of Numerically Parallel Divisors*, arXiv:2207.14432.
- [11] R. LAZARSFELD, *Positivity in Algebraic Geometry. I. Classical setting: line bundles and linear series*. Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. A Series of Modern Surveys in Mathematics vol. 48. Springer-Verlag, Berlin, 2004.
- [12] A. LEVIN, *Great common divisors and Vojta's conjecture for blowup of algebraic tori*, Invent. Math. **215** (2019), no. 2, 493–533.
- [13] A. LEVIN AND J. T.-Y. WANG, *Greatest common divisors of analytic functions and nevanlinna theory on algebraic tori*, J. Reine Angew. Math., **767**(2020), 77–107.
- [14] J. NOGUCHI AND J. WINKELMANN, *Nevanlinna Theory in Several Complex Variables and Diophantine Approximation*, Grundlehren Math. Wiss., 350, Springer-Verlag, Tokyo, 2014.
- [15] J. NOGUCHI, J. WINKELMANN AND K. YAMANOI, *The second main theorem for holomorphic curves into semi-Abelian varieties*, Acta Math., **188**(2002), no. 1, 129–161.
- [16] M. PĂUN AND N. SIBONY, *Value distribution theory for parabolic Riemann surfaces*, Panor. Synthèses, 56[Panoramas and Synthèses], Société Mathématique de France, Paris, 2021, 13–72.
- [17] M. RU, *Nevanlinna Theory and its Relation to Diophantine Approximation*, World Scientific, 2021. Publishing Co., Inc., River Edge, NJ, 2021.
- [18] M. RU AND W. STOLL, *The second main theorem for moving targets*, Journal of Geometric Analysis, **1**(1991), 99–138.
- [19] M. RU AND J. T.-Y. WANG, *Truncated second main theorem with moving targets*, Trans. Amer. Math. Soc. **356** (2004), no. 2, 557–571.

- [20] M. RU AND J. T.-Y. WANG, *Defect relation of $n + 1$ components through the GCD method*, Pacific Journal of Mathematics, **338** (2), 2025, 349–371.
- [21] M. RU AND P. VOJTA, *Birational Nevanlinna constant and its consequences*, Amer. J. Math., **142** (2020), no. 3, 957–991.
- [22] J. H. SILVERMAN, *Generalized greatest common divisors, divisibility sequences, and Vojta’s conjecture for blowups*, Monatsh. Math. **145** (2005), 333–350.
- [23] W. STOLL, *Value Distribution on Parabolic Spaces*, Lecture Notes in Mathematics, Vol. 600. Springer-Verlag, Berlin-New York, 1977.
- [24] J. T.-Y. WANG, *An effective Roth’s theorem for function fields*, Rocky Mountain J. Math. **26**(1996), no. 3, 1225–1234.
- [25] J. T.-Y. WANG AND Y. YASUFUKU, *Generalized greatest common divisors of integral points of numerically equivalent divisors*, Algebra & Number Theory **15** (2021), no. 1, 287–305.
- [26] J. T.-Y. WANG, *Cartan’s conjecture with moving targets of same growth and effective Wirsing’s theorem over function fields*, Math. Z. **234** (2000), 739–754.
- [27] P. VOJTA, *Diophantine approximation and Nevanlinna theory*, in Arithmetic geometry, Lecture Notes in Math. 2009, Springer, Berlin, 2011, 111–224.

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